

Thomas Jefferson **Physics Olympiad**

A high school physics contest

This exam covers concepts related to electromagnetism and superconductors and explores current insights into superconductor research and applications.

TJ Physics Team <tjhsstphysicsteam@gmail.com>

Electromagnetism and Superconductors

If you have any questions or concerns, please email TJ Physics Team.

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Rules

Teams will have a total of four days to complete the TJPhO. All teams are required to submit their response with a cover page listing the team name and number, team member names, team email, and the date. Each submitted page should also have the problem number. All other formatting decisions are delegated to the teams themselves, with no one style favored over another. We suggest that teams use $\text{\LaTeX}$ for typesetting. The easiest way we've found to typeset $\text{\LaTeX}$ is with Overleaf.

There will be two types of problems on the online exam:

- **Background:** Background problems will test knowledge of classical physics and involve higher-level conceptual thinking and application of fundamental formulas and calculations.
- **Applied:** Applied problems will explore the applications of superconductors, use known concepts to generalize to higher systems, and combine knowledge from a variety of different topics.

Even if you are not able to fully solve a problem, please write up what you can. We want to see how you approached the problem, and partial credit may be awarded.

Collaboration

Students participating in the competition may only correspond with other members of their team. No other correspondence is allowed, including mentors, teachers, professors, and other students. Teams are not allowed to use online/print resources or post content on online forums asking questions related to the exam. We welcome teams to email us if there are any questions or concerns. Teams may use any computational resources they might find helpful, such as Wolfram Alpha/Mathematica, Matlab, Excel, or programming languages (C++, Java, Python, etc).

Submission

Teams must submit their solutions by email to tjhsstphysicsteam@gmail.com by the deadline 11:59 PM EST on April 10th. Late submissions will not be considered. Solutions should be written in English and submitted as a single PDF document with the .pdf extension. The email must contain "Submission" in the subject line. Only one person per team, the person who registered through the Google Form, should send the final submission.

Awards

The top five teams will receive awards.

1st place: \$200, 3 electronic copies of *Superconductivity: A Very Short Introduction* by Stephen J. Blundell, 3 one-year subscriptions to WolframAlpha Notebook Edition

2nd place: \$150, 3 electronic copies of *Superconductivity: A Very Short Introduction* by Stephen J. Blundell, 3 one-year subscriptions to WolframAlpha Notebook Edition

3rd place: \$100, 3 electronic copies of *Superconductivity: A Very Short Introduction* by Stephen J. Blundell, 3 one-year subscriptions to WolframAlpha Notebook Edition

4/5th place: 3 electronic copies of *Superconductivity: A Very Short Introduction* by Stephen J. Blundell, 3 one-year subscriptions to WolframAlpha Notebook Edition

Sponsors

We are incredibly grateful to our sponsors, Dr. Adam Smith and Mr. Robert Culbertson, and Thomas Jefferson High School for Science and Technology for their support. We would also like thank our sponsors WolframAlpha, the Thomas Jefferson Partnership Fund, and MinutePhysics for making this Olympiad possible.



Introduction

In 1911, Dutch physicist Heike Kamerlingh Onnes was experimenting with low-temperature mercury when he noticed a strange property — when he cooled the mercury to the same temperature as liquid helium, the resistance completely disappeared. This new phenomenon called superconductivity emerged as an entire field of research. For the past century, physicists have been on the race to create substances that exhibit superconductivity at lower pressures and higher temperatures. Just this past October, superconductivity was discovered in a compound consisting of carbon, sulfur and hydrogen at a record-high temperature of 15°C . These breakthroughs in superconductivity open the door for physicists and engineers alike to develop advanced and impactful technologies.

The TJ Physics Team is proud to announce the third annual TJ Physics Olympiad (TJPhO)! We hope this contest will introduce you to the fascinating world of superconductivity and provide a good starting point to explore electromagnetism. As per the nature of superconductivity, there will be quite a bit of electromagnetism involved in solving certain problems. To accommodate those that may not be familiar with these subjects, we have included an overview of multivariable calculus and background problems to build an understanding of electromagnetism.

This contest covers a wide range in superconductor research, from the underlying principles to the current applications such as MRIs, quantum computers, and particle accelerators. We hope you have as much fun learning about superconductors as we had writing this contest.

Happy exploring!

TJ Physics Team Officers

Multivariable Calculus

This section is dedicated to the conceptual background of multivariable calculus necessary for the exam. You do not need to mathematically understand the topics in this section, as all problems may be done using single-variable calculus.

For a function $f(x, y)$, the partial derivative of f with respect to x is given by

$$\frac{\partial f}{\partial x} = \lim_{u \rightarrow x} \frac{f(u, y) - f(x, y)}{u - x}.$$

The idea is that all other variables except the variable that the derivative is being taken with respect to is held constant.

A vector field $\mathbf{F}(x, y, z)$ is a function that outputs a vector at every point in space. The del operator $\nabla = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$ is the vector differential operator used to define the divergence and curl. The divergence of a vector field $\nabla \cdot \mathbf{F}$ represents how much a vector field acts like a “source” or a “sink” at a given point. The curl of a vector field $\nabla \times \mathbf{F}$ represents the vector “circulation” at a certain point. One may also think of the divergence and curl as the dot and cross product between the del operator and $\mathbf{F}$ respectively. You do not need to calculate divergence and curl for this exam.

Line integrals are used to integrate a vector field $\mathbf{F}$ over a curve C , parameterized by $\mathbf{r}(t)$. You may have seen this when computing the work done to an object along a path. The line integral essentially takes infinitesimal steps along C , taking the dot product of the displacement vector with the vector field at each step. A line integral starting at point A and ending at point B is denoted by

$$\int_A^B \mathbf{F} \cdot d\mathbf{r}.$$

If C is a closed loop, the line integral is denoted by

$$\oint_C \mathbf{F} \cdot d\mathbf{r}.$$

Surface integrals are used to integrate a vector field $\mathbf{F}$ across a surface S by determining the flux of the field through the surface, or how much of $\mathbf{F}$ is going through it. This is done by taking a dot product of $\mathbf{F}$ with a normal vector to S at every point on the surface, and then integrating over both parameters. Essentially, the surface integral adds up all of the contributions of the vector field that go out of the surface, multiplied by the area of each small section of the surface. A surface integral is denoted by

$$\iint_S \mathbf{F} \cdot d\mathbf{S}.$$

Note that line and surface integrals are difficult to calculate in general. All integrals of these two types on the exam may be simplified to a single-variable integral.

1 Background Problems

Electric and Magnetic Fields (20 pts.)

1

Charged particles, such as protons or electrons, will produce an electric field $\mathbf{E}$. By Coulomb's Law, the electric field of a particle with charge q at a position $\mathbf{r}$ relative to the particle is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}},$$

where ϵ_0 is the permittivity of free space, with $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \cdot \text{N}^{-1} \cdot \text{m}^{-2}$. Electric current is the net charge flowing through a region in space per unit time. Moving charges will produce a magnetic field $\mathbf{B}$. By Ampere's Law, the line integral of $\mathbf{B}$ around a closed loop C is proportional to the current passing through C .

$$\oint_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 I,$$

where μ_0 is the permeability of free space, with $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m} \cdot \text{A}^{-1}$.

The force on a particle with charge q moving with velocity $\mathbf{v}$ through an electric and magnetic field is given by the Lorentz force $\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$.

- a) Two charges $-q$ and $2q$ are at $x = -a$ and $x = a$ respectively. A test charge is placed at $x = d$. Find d if the force on the test charge is zero.
- b) Suppose an infinite straight wire has constant current I . Taking C to be a circular loop with radius r and appealing to symmetry, use Ampere's Law to determine the magnitude of the magnetic field at a distance r from the wire.
- c) An electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ are directed along the $+z$ -axis. A mass m with charge q starts at the origin with velocity v along the $+x$ axis. Find the position of the particle as a function of time and describe the motion.

The Lorentz equation and Maxwell's equations form the basis of classical electrodynamics. According Maxwell's equations, time dependent electric and magnetic fields in a vacuum will interact such that they propagate with speed $c = \frac{1}{\sqrt{\epsilon_0\mu_0}} = 3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$.

Conductors (20 pts.)

2

The electric potential V is defined as potential energy per unit charge. The electric potential difference ΔV is the work per unit charge needed to move a particle from point A to point B along any path.

$$\Delta V = V_B - V_A = - \int_A^B \mathbf{E} \cdot d\boldsymbol{\ell}.$$

A conductor allows charged particles to move freely within it. The ability to conduct electrical current is characterized by the conductivity σ . For a conductor with cross-sectional area A , Ohm's Law states that an electric field $\mathbf{E}$ will produce a current density $\mathbf{J} = \sigma\mathbf{E}$, where $J = \frac{I}{A}$. For a resistor with resistance R , an alternate and perhaps more familiar version of Ohm's Law states $I = \frac{V}{R}$, where V is the potential difference across the resistor.

- a) Find the resistance of a conductor with cross-sectional area A , length L , and conductivity σ in terms of the given variables.

Even though there exists a net current flow for a conductor in an electric field, the electrons are moving randomly, which causes scattering. Electron scattering produces an drag force $F_{drag} = \frac{mv}{\tau}$, where m is the mass of the electron and τ is the average time between collisions.

- b) After a long time, the electrons in a current will move with constant drift velocity v_d . Find v_d for a conductor placed in a uniform electric field $\mathbf{E}$.
- c) Suppose the conductor has electron density n and conductivity σ . Show the average time between electron collisions is $\tau = \frac{m\sigma}{e^2n}$.

Below a critical temperature, the resistance of an ideal superconductor is zero, allowing superconductors to carry current for years without noticeably decaying. This effect is known as supercurrent.

Circuits (30 pts.)

3

Voltage drop is defined as the electric potential difference between two points in a circuit. Kirchhoff's loop rule states that the sum of the electric potential differences in any closed circuit loop is zero.

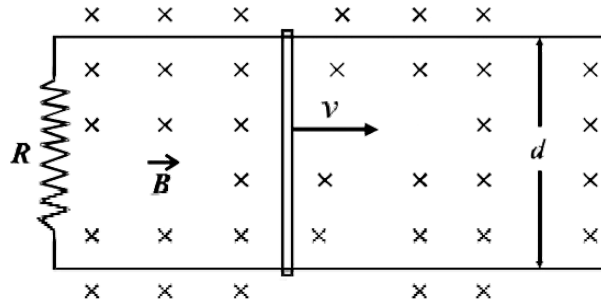
- a) Two resistors with resistances R_1 and R_2 are connected in series across a battery. The circuit may be reduced to a battery and a single resistor with resistance R_{eq} . Using Kirchhoff's loop rule, find R_{eq} in terms of R_1 and R_2 . Determine R_{eq} in the limit as $R_2 \rightarrow 0$.
- b) Repeat part a) with the two resistors connected in parallel.

Magnetic flux Φ_B is defined as the total magnetic field passing through a surface. For a magnetic field $\mathbf{B}$ and surface S ,

$$\Phi_B = \iint_S \mathbf{B} \cdot d\mathbf{S}.$$

The Faraday-Lenz law states that a time dependent magnetic flux through the area bounded by a circuit will induce an opposing electromotive force (EMF) $\mathcal{E} = -\frac{d\Phi_B}{dt}$, with positive direction determined by the right hand rule. The EMF may be thought of as an electric potential difference, or voltage from a battery.

A freely moving conducting rod and a resistor with resistance R are placed across two parallel conducting rails separated by a distance d . The rod moves with constant velocity v to the right, perpendicular to a uniform constant magnetic field B directed into the page as shown in the figure below.



- c) Find the magnitude and direction of the induced EMF.
- d) Let the linear density of electrons in the rod be λ . Find the force from the magnetic field on the charges in the freely moving conducting rod.
- e) Joule's first law states the resistor will dissipate power $P = IV = I^2R$ in the form of heat. By conversation of energy, and equal power must be delivered to the circuit. Where does this power come from?

Superconductors have gained attention for their ability to effectively eliminate resistive Joule heating, thereby carrying supercurrents and enabling perfect transfer of energy.

Diamagnetism (40 pts.)

4

Let $\hat{n}$ be the unit vector normal to the plane of a current loop, with positive direction determined by the right hand rule. The magnetic moment $\mathbf{m}$ of a circular loop of radius r carrying current I is $I\pi r^2\hat{n}$. The orbital motion of electrons will form tiny current loops with a permanent magnetic moment.

An applied magnetic field in a material will cause the permanent magnetic moments to change depending on the magnetization $\mathbf{M}$, which is defined as the net magnetic moment per unit volume. When a magnetic field $\mathbf{B}_0$ is applied to a material with magnetization $\mathbf{M}$ parallel to $\mathbf{B}_0$, the internal magnetic field is

$$\mathbf{B} = \mathbf{B}_0 + \mu_0\mathbf{M} = \mathbf{B}_0(1 + \chi_m),$$

where χ_m is the magnetic susceptibility. Diamagnetism occurs when the applied magnetic field induces magnetic moment in the opposite direction of $\mathbf{B}_0$.

To understand why diamagnetism occurs, consider N uniformly spaced electrons, each moving with speed v in a circle of radius r . A uniform magnetic field $\mathbf{B}_0$ is then applied. Suppose half the electrons magnetic moments are parallel to $\mathbf{B}_0$ and half are antiparallel. Let m and e be the mass and charge of the electron respectively.

- a) Working with the assumption $B_0 \ll \frac{mv}{re}$, find the change in speed Δv of the electrons due to the applied magnetic field. Assume the orbit of the electrons do not change.
- b) Determine the magnetic moment of a single electron whose moment is parallel to $\mathbf{B}_0$. Then summing over all electrons, determine the net magnetic moment.
- c) Find χ_m in terms of the electron density ρ , r , and fundamental constants.

Due to electromagnetic induction on the atomic scale, all materials exhibit diamagnetism. However, the effects may be masked by other stronger properties such as ferromagnetism, which is responsible for the magnetic properties of typical magnets (e.g. iron, nickel).

Magnetic Dipoles (40 pts.)

5

In spherical coordinates, every point is described by r , θ , and ϕ , representing the radial distance from the origin, the azimuthal angle from the $+z$ -axis, and the polar angle in the xy -plane respectively. To check your understanding, it may help to draw a diagram and show the following conversions,

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$$

A magnetic dipole produces a magnetic field similar to that of a north and south pole pair. The strength of the dipole is the magnetic moment, which points from south to north. For a dipole placed at the origin with dipole moment $\mathbf{m}$ along the $+z$ -axis, the magnetic field in spherical coordinates is given by

$$B_r = \frac{\mu_0 m}{2\pi r^3} \cos \theta, \quad B_\theta = \frac{\mu_0 m}{4\pi r^3} \sin \theta, \quad B_\phi = 0.$$

where B_r , B_θ , and B_ϕ are the components of the magnetic field along the directions of increasing r , θ , and ϕ respectively.

- a) A magnetic dipole with moment $\mathbf{m}$ is placed in a uniform magnetic field $\mathbf{B}_0$ directed opposite to the dipole moment. Find R for which no magnetic field lines pass through the shell of with radius R centered at the dipole.

The magnetic dipole is replaced with a homogeneous conducting sphere with radius R , surface charge density σ , and identical magnetic moment. The external magnetic field produced is the same as that of the removed dipole. Internally, the magnetic field produced is $\frac{2}{3}\mu_0\mathbf{M}$, where $\mathbf{M}$ is the magnetization of the conductor.

- b) Find the B_r , B_θ , and B_ϕ components of the total magnetic field as a function of r and θ .
- c) Find the speed of the charge carriers moving on the surface of the conductor.

With the right current distribution on the shell of radius R , it is possible to eliminate magnetic fields in the interior. The ability for superconductors to expel applied magnetic fields is explored in greater depth via the London equations and the Meissner effect.

Meissner Effect (50 pts.)

6

Superconductors exhibit a phenomenon called the Meissner effect, wherein the superconductor repels all external magnetic fields. The effect explains why placing a superconductor on top of a magnet will cause it to levitate.

To treat more complex systems, consider the local differential forms of the Faraday-Lenz law and Ampere's law, which respectively state

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

By the results from problem 2 and Ohm's law, the current density $\mathbf{J} = \frac{ne^2\tau}{m}\mathbf{E}$. Taking the time derivative of both sides gives the first London equation,

$$\frac{d\mathbf{J}}{dt} = \frac{d(n\tau)}{dt} \frac{e^2}{m} \mathbf{E} = \frac{n_s e^2}{m} \mathbf{E},$$

where the number density $n_s = \frac{d(n\tau)}{dt}$.

- a) The Helmholtz equation for the magnetic field states

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda^2} \mathbf{B}.$$

where the Laplacian operator $\nabla^2 = \nabla \cdot \nabla$ is the divergence of the gradient. Use the first London equation to determine the penetration depth λ in terms of m , μ_0 , n_s , and e .

Start by taking the curl of Ampere's law. Then, apply Faraday's law and integrate with respect to time. You are given the magnetic field identity

$$\nabla \times \nabla \times \mathbf{B} = -\nabla^2 \mathbf{B}.$$

- b) Suppose the magnetic field within the superconductor only varies in one dimension, meaning the operator ∇ is $\frac{d}{dx}$. For a certain superconductor, the magnetic field 500 nm deep into the superconductor is 5 times weaker than the magnetic field at the surface. Using the Helmholtz equation, find the penetration depth to the nearest nanometer.

Your analysis in part b) should mathematically justify the Meissner effect. The explanation behind how the effect works is given by the second London equation,

$$\nabla \times \mathbf{J} = -\frac{n_s e^2}{m} \mathbf{B},$$

derived by taking the curl of the first London equation and applying Faraday's law. The equation states that an external magnetic field induces a current in the superconductor, which produces an internal magnetic field that cancels the external field, hence the Meissner effect.

- c) Two isolated conductors sit far apart at room temperature. Conductor A has a small external magnetic field applied to it, while conductor B does not. They are both cooled down below their superconducting critical temperatures. An identical small external magnetic field is then applied to B. Using conclusions from the London equations, describe what happens to the external and internal magnetic fields for each superconductor.
- d) Now the external magnetic fields are turned off for A and B. Describe what happens to the external and internal magnetic fields for each superconductor.
- e) Suppose that A is a perfect conductor instead of a superconductor. Repeat part d) for the perfect conductor by describing the new behavior of the external and internal magnetic fields.

The Meissner effect is a special and unexpected property of superconductors, different from perfect conductivity or perfect diamagnetism. The effect also manifests differently in type I and II superconductors, with the latter forming magnetic flux tubes rather than displaying complete Meissner effects.

2 Applied Problems

Cooling Superconductors (50 pts.)

7

A refrigerator may be used to cool a superconductor to the critical temperature T_c by absorbing heat Q_c from a cold reservoir and releasing heat Q_h to a hot reservoir. Because this process reverses heat flow, it requires an external work W . A refrigerator operates optimally when the total entropy, a measure of the disorder in the system, stays constant. For a system at constant temperature T , the change in entropy is $\Delta S = \frac{Q}{T}$, where Q is the heat transferred to the system.

- a) The coefficient of performance (COP) of a refrigerator is $\frac{Q_h}{W}$. Find the COP of an optimal refrigerator operating between cold and hot reservoirs at temperatures T_c and T_h . Express your answer in terms of T_c and T_h .

Specific heat c is defined as the heat per unit mass required to increase temperature by one kelvin. By the equipartition theorem, the energy per mole associated with each molecular degree of freedom is $\frac{1}{2}kTN_A$, where $k = 1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1}$ is the Boltzmann constant and $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ is the Avogadro constant.

- b) Considering translational, rotational, and vibrational degrees of freedom, calculate the specific heat of a solid with molar mass M . Express your answer in terms of T , M , and fundamental constants.

The Debye model corrects the equipartition theorem at low temperatures by treating molecules as quantum harmonic oscillations with discrete energies. With the Debye model, the energy per mole for a solid with Debye temperature T_D is

$$\frac{9kT^4N_A}{T_D^3} \int_0^{T_D/T} \frac{x^3}{e^x - 1} dx.$$

- c) Find the Debye specific heat in the high temperature limit $T \gg T_D$. Explain why this answer is expected.
- d) Find the Debye specific heat in the low temperature limit $T \ll T_D$. You are given the following integral.

$$\int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}.$$

Suppose a superconductor has mass m and molar mass M . A refrigerator operating optimally between the superconductor and a hot reservoir with constant temperature T_h is used to cool the superconductor from an initial temperature $T_i = T_h$ to T_c .

- e) Find the work W required and the change in entropy ΔS of the superconductor in terms of m , M , T_i , T_c , T_D , and fundamental constants. Comment on the disorder in the superconductor as it is cooled. Assume the low temperature limit for the Debye specific heat holds during the entire process.
- f) Using your expressions in part e), calculate W and ΔS if 50 g of tin initially at 100 K is cooled to the critical temperature of 7.2 K. For tin, $T_D = 200 \text{ K}$ and the molar mass is $119 \text{ g} \cdot \text{mol}^{-1}$.

The cost of cooling superconductors to a few kelvin using liquid helium often offsets the advantages of eliminating Joule heating. As a result, research is centered around high-temperature superconductors, which have critical temperatures above 77 K, the boiling point of liquid nitrogen. In October 2020, scientists at the University of Rochester in New York discovered the first room-temperature superconductor, with a critical temperature of 288 K under 267 gigapascals of pressure.

Magnetic Resonance Imaging (MRI) Machines (40 pts.)

8

An inductor is a cylindrical coil of wire, shaped similarly to a slinky, that produces a constant magnetic field $\mathbf{B}$ oriented parallel to the length inside the solenoid and no magnetic field outside. The inductance L of an inductor gives the tendency to resist a time-varying current, which produces an opposing electromotive force $\mathcal{E} = -L \frac{dI}{dt}$. For an inductor with cross-sectional area A , length ℓ , and n turns per unit length, $L = \mu_0 n^2 A \ell$.

- a) Using Ampere's Law with a rectangular loop, find the magnetic field B inside the inductor in terms of L , A , ℓ , I , and fundamental constants.

An MRI machine uses a large superconducting inductor with inductance L and n turns per unit length to produce a strong magnetic field. The inductor is connected in parallel with a persistence resistor with resistance R_P and a DC current source until the final magnetic field strength B_f is achieved. The current source is then disconnected and the resistor is cooled until it becomes superconducting. The superconducting loop then carries the current in persistence mode. Suppose due to impurities and incomplete Meissner effects, the inductor and resistor both have a small but non-zero resistance $R \ll R_P$ when below the supercritical temperature.

- b) Suppose the current source delivers a constant current I . Find $B(t)$ in terms of L , n , R_P , $\eta = \frac{R}{R_P}$, and I . Determine B_{max} , the maximum value of $B(t)$. How does B_{max} vary with η ?
- c) Find $B_P(t)$, where t is the time after transitioning to persistence mode, in terms of B_f , R , and L .
- d) Find R if the magnetic field of a 1.5 T MRI machine with diameter 0.8 m, length 2.5 m, and initial current 500 A decays by 1 mT after one year. Express your answer in scientific notation rounded to the nearest hundredth.
- e) In persistence mode, quenching may occur if the resistor is suddenly heated and loses superconducting properties. Calculate the heat Q released through quenching using the numeric values given in part d).

Superconducting Qubits (50 pts.)

9

A capacitor is a circuit device composed of two parallel oppositely charged plates that stores energy in an electric field. The voltage drop across the capacitor is $V = \frac{Q}{C}$, where Q is the charge on the capacitor plates and C is the capacitance.

An LC circuit composed of a capacitor and inductor serves as the foundation of the superconducting qubit. Suppose the electric and magnetic field energies are analogous to kinetic and potential energies respectively.

- a) Find the total energy in an LC circuit.
- b) The single particle quantum harmonic oscillator with potential function $\frac{1}{2}kx^2$ has allowed energy values $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$, where $\omega = \sqrt{\frac{k}{m}}$ and n is a non-negative integer. Drawing on some familiarity with the kinetic and potential energy of a particle, find the allowed energy values for a quantum LC circuit.
- c) Why are the allowed energies in a quantum LC circuit problematic for the implementation of a two energy level qubit?

A Josephson junction is composed of two superconductors separated by a thin insulating wafer and carries a supercurrent via quantum tunnelling. The current and voltage drop across the junction are given by the Josephson equations,

$$I = I_c \sin(\phi), \quad V = \frac{\hbar}{2e} \frac{d\phi}{dt}.$$

where I_c is the critical current and ϕ is the phase difference across the insulator. Suppose the inductor in an LC circuit is replaced by a Josephson junction.

- d) Find the potential energy function of the superconducting qubit. Explain why the addition of the junction solves the problem identified in part c).
- e) Calculate the maximum temperature for a useful superconducting qubit operating at 8 GHz to the nearest mK.

Quantum computing initiatives at Google, IBM, Rigetti, etc. use superconducting qubits to create quantum devices. In October 2019, scientists at Google announced their 53-qubit quantum computer surpassed the capabilities of the best classical supercomputers.

Particle Accelerators (60 pts.)

10

The Large Hadron Collider at CERN uses superconducting magnets to accelerate protons to relativistic speeds. The protons have mass $m = 1.67 \times 10^{-27}$ kg and are injected into a circular tube with radius $r = 27$ km. Two equal and opposite currents with magnitude I are sent through two circular superconducting wires with radii $r \pm d$, with $d = 2$ cm. The superconductors have a small resistance R and create a magnetic field perpendicular to the velocity of the protons. Suppose the protons are injected with zero initial speed and the current is increased at a constant rate α , which accelerates the protons to a final speed $v = \beta c$, where $\beta = 1 - 10^{-4}$.

The radius of motion of a particle in a magnetic field $\mathbf{B}$ perpendicular to velocity is given by the gyroradius $r_g = \frac{\gamma m v}{|q|B}$, where $\gamma = \frac{1}{\sqrt{1-\beta^2}}$. An accelerating charged particle will emit radiation in the form of electromagnetic waves. By the Larmor formula, the power radiated is

$$P = \frac{q^2 a^2}{6\pi\epsilon_0 c^3},$$

where the acceleration a is equal to $\frac{\gamma}{m} \left| \frac{d\mathbf{P}}{dt} \right|$. The relativistic momentum four-vector $\mathbf{P} = (\gamma mc, \gamma m \mathbf{v})$ satisfies

$$|\mathbf{P}|^2 = (\gamma m v)^2 - (\gamma m c)^2 = \gamma^2 m^2 c^2 \left(\frac{v^2}{c^2} - 1 \right) = -m^2 c^2.$$

- a) If the protons are kept at the final speed, find the power dissipated by the superconducting wires. Express your answer as R times a constant in scientific notation to the nearest hundredth.
- b) Find the total heat dissipated by the superconducting wires as the protons are accelerated to the final speed. Express your answer as $\frac{R}{\alpha}$ times a constant in scientific notation to the nearest hundredth.
- c) Find the minimum external power needed to accelerate one proton to the final speed. Express your answer symbolically using the given variables.

Superconducting magnets enable particle accelerators to achieve faster relativistic speeds. For comparison, the first particle accelerators used a voltage difference to accelerate protons to 1 MeV, whereas the LHC accelerates protons to 6.5 TeV.

3 BCS Theory

Bardeen-Cooper-Schrieffer (BCS) theory proposed the first microscopic treatment of superconductors. This section is dedicated to the foundations of BCS theory, culminating with the derivation of Cooper pairs.

Quantum Statistics (75 pts.)

11

In quantum mechanics, the wave function $\psi(\mathbf{r})$ is a complex-valued function that describes the state of a system. The magnitude of $\psi(\mathbf{r})$ is the probability amplitude, meaning the particle is in the interval $(\mathbf{r}_0, \mathbf{r}_0 + d\mathbf{r})$ with probability $|\psi(\mathbf{r}_0)|^2 d^3\mathbf{r}$, where $d^3\mathbf{r} = dr_x dr_y dr_z$.

Two particles of the same species are indistinguishable in quantum mechanics. For a two particle wave function $\psi(\mathbf{r}_1, \mathbf{r}_2)$, switching the position vectors will not change the probability amplitude. According to the spin-statistics theorem,

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \psi(\mathbf{r}_2, \mathbf{r}_1) \text{ for bosons, } \psi(\mathbf{r}_1, \mathbf{r}_2) = -\psi(\mathbf{r}_2, \mathbf{r}_1) \text{ for fermions.}$$

- a) Using the spin-statistics theorem, show that bosons have integral spin while fermions have half-odd-integral spin. You may assume both species of particles exist and that particles with the same spin are either all bosons or all fermions.
- b) Justify the Pauli exclusion principle, which states that two fermions can not occupy the same quantum state.
- c) At low temperatures around one kelvin, liquid ^4He is a superfluid, flowing with zero viscosity and no kinetic energy losses, whereas ^3He is a liquid. What does this indicate about the electrons in a superconducting metal?

The Schrödinger equation determines the properties of a wave function $\psi(\mathbf{r})$. In three dimensions, the time independent equation is

$$-\frac{\hbar^2}{2m}\nabla_{\mathbf{r}}^2\psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r}),$$

where $\nabla_{\mathbf{r}}^2 = \frac{\partial^2}{\partial \mathbf{r}^2}$. The terms $-\frac{\hbar^2}{2m}\nabla_{\mathbf{r}}^2$ and $V(\mathbf{r})$ are the kinetic and potential energies of the state respectively.

Suppose the conduction electrons in a metal are modelled as a free Fermi gas, non-interacting fermions with $V(\mathbf{r}) = 0$. Furthermore, impose the condition that the wave function repeats with period L along the x , y , and z directions.

The energy density of states per unit volume $D(E)$ is defined such that the number of states with energy between E and $E + dE$ is $\Omega D(E)dE$, where Ω is the volume of the system.

- d) Let the mass of the electron be m . Find all values of the momentum $\mathbf{p}$ which satisfy the Schrödinger equation and the imposed periodic boundary conditions.
- e) Find $D(E)$ in terms of m , E and fundamental constants. It may help to plot values of $\mathbf{p}$ and partition the space into homogeneous volume elements.
- f) Suppose $F(\mathbf{p})$ is a function of $\mathbf{p}$. Find the constant α for which

$$\sum_{\mathbf{p}} F(\mathbf{p}) = \frac{\Omega}{\alpha} \int F(\mathbf{p}) d^3\mathbf{p}$$

where $\mathbf{p}$ is summed over values found in part d) and Ω is the volume of the system. Although this result is derived for large cubic volumes, it holds for large systems of arbitrary shape.

The energy density of states per unit volume is useful in dealing with many-particle systems. Let $N \gg 1$ be the total number of particles in a boson gas with chemical potential μ as the thermodynamic conjugate variable. Using the grand canonical ensemble and Bose-Einstein statistics, the average number of particles in state i at temperature T is

$$\langle n_i(\mu, T) \rangle = \frac{1}{e^{(\varepsilon_i - \mu)/kT} - 1},$$

where ε_i is the energy of state i and μ is the chemical potential. Because bosons do not follow the Pauli exclusion principle, below a critical temperature T_c , the bosons will begin to simultaneously occupy the ground state with $\varepsilon_0 = 0$ while the rest occupy excited states with $\mu = 0$.

- g) Find all possible values of $\langle n_i(\mu, T) \rangle$ for bosons and fermions. Then express $\langle n_0(T) \rangle$ for bosons in terms of N , T , and T_c . Verify your answer in the limiting cases. Why is your expression for $\langle n_0(T) \rangle$ problematic when comparing superfluids and superconductors?

Working in the grand canonical ensemble, there are certain subtleties that must be taken into account when considering superfluids and superconductors. Most notably, superfluids will exhibit spontaneously broken symmetry, while superconductors have a different property called topological order.

Cooper Pairs (75 pts.)

12

Superconductivity is due to the formation of Cooper pairs, pairs of electrons that form a bound state with an attractive potential. Despite electrons having the same charge, the interaction is attractive due to quantum effects. As a simplified classical explanation, consider a slowly moving electron which pulls positive ions in the surrounding lattice towards it. This creates a small region of positive charge, attracting another electron.

Consider two electrons at positions $\mathbf{r}_1$ and $\mathbf{r}_2$ interacting via a potential $V(\mathbf{r}_1 - \mathbf{r}_2)$. Let $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ be the relative position and $\mathbf{R} = \frac{\mathbf{r}_1 + \mathbf{r}_2}{2}$ be the center of mass.

- a) Write the Schrödinger equation for the wave function $\psi(\mathbf{r}_1, \mathbf{r}_2)$. Then rewrite $\psi(\mathbf{r}_1, \mathbf{r}_2)$ as the wave function $\psi(\mathbf{r}, \mathbf{R})$ by expressing the Laplacian in terms of $\mathbf{r}$ and $\mathbf{R}$ using the chain rule for multivariate functions,

$$\frac{\partial}{\partial \mathbf{r}_a} = \frac{\partial \mathbf{r}}{\partial \mathbf{r}_a} \frac{\partial}{\partial \mathbf{r}} + \frac{\partial \mathbf{R}}{\partial \mathbf{r}_a} \frac{\partial}{\partial \mathbf{R}}.$$

- b) Since the potential does not depend on $\mathbf{R}$, the wave function $\psi(\mathbf{r}, \mathbf{R})$ is of the form $\psi(\mathbf{r})e^{-i\mathbf{P}\cdot\mathbf{R}/\hbar}$. Simplify the expression you obtained in part a) and then set $P = 0$.

The wave function $\psi(\mathbf{r})$ is expressed in position space. There exists an wave function $\phi(\mathbf{p})$ in momentum space equivalent to $\psi(\mathbf{r})$. The Fourier transform $\mathcal{F}$ and inverse Fourier transform relates $\psi(\mathbf{r})$ and $\phi(\mathbf{p})$ by the equations

$$\begin{aligned}\phi(\mathbf{p}) &= \mathcal{F}[\psi(\mathbf{r})] = \int_{-\infty}^{\infty} \psi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} d^3\mathbf{r} \\ \psi(\mathbf{r}) &= \mathcal{F}^{-1}[\phi(\mathbf{p})] = \frac{1}{(2\pi\hbar)^3} \int_{-\infty}^{\infty} \phi(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} d^3\mathbf{p},\end{aligned}$$

- c) Apply the Fourier transform to show that the Schrödinger equation for two interacting electrons in momentum space is

$$\frac{\Omega}{(2\pi\hbar)^3} \int V(\mathbf{p} - \mathbf{p}') \phi(\mathbf{p}') d^3\mathbf{p}' = (E - 2\varepsilon_{\mathbf{p}}) \phi(\mathbf{p}),$$

where $V(\mathbf{p} - \mathbf{p}')$ is the normalized potential function operator in momentum space. Ω is the volume of the system and $\varepsilon_{\mathbf{p}} = \frac{p^2}{2m}$.

The electrons in a superconducting material will experience an attractive interaction if their energies are above the Fermi energy threshold $\varepsilon_F \gg \hbar\omega_D$, where ω_D is the Debye frequency. ε_F is the maximum energy of an electron at absolute zero, and ω_D is the frequency of the phonons that mediate the electron-electron interaction.

To simplify the problem, assume the attractive potential is given by

$$V(\mathbf{p} - \mathbf{p}') = \begin{cases} -V_0 & \varepsilon_{\mathbf{p}} - \varepsilon_F, \varepsilon_{\mathbf{p}'} - \varepsilon_F < \hbar\omega_D \\ 0 & \text{otherwise} \end{cases}$$

where V_0 is a constant.

- d) Briefly describe the qualitative properties of the attractive potential $V(\mathbf{r})$.
e) Find the function $f(\mathbf{p})$ for which

$$\Omega V_0 \int f(\mathbf{p}) d^3\mathbf{p} = 1.$$

- f) Suppose the interaction is weak with $V_0 \ll \frac{\beta}{\Omega\sqrt{\varepsilon_F}}$, where β is a small constant with appropriate units. Write the integral

$$\int f(\mathbf{p}) d^3\mathbf{p}$$

as an integral over energy

$$\int g(\varepsilon) d\varepsilon.$$

- g) Find $2\varepsilon_F - E$ in terms of m , V_0 , ε_F , ω_D , Ω , and fundamental constants. How does $2\varepsilon_F - E$ behave as $V_0 \rightarrow 0$? Why does the relationship show the existence of Cooper pairs?

4 Current Research

Here is an enumeration of articles and web pages on current superconductor research, for those interested. Click on the hyperlinks for more information.

Properties

- Fe-based Superconductors
- Graphene Superconductors
- Low Pressure Superconductors
- Room Temperature Superconductors

Applications

- Chip Interconnects
- Energy Storage Rings
- Generators with Superconducting Wires in Rotors
- IR Sensors
- Josephson Junctions
- Magnetic Bearings
- Magnetic Separation
- Magnets for Fusion Devices
- Magnets for Magnetically Levitated Trains (MAGLEV)
- Magnets for Magnetic Resonance Imaging
- Power Transmission Lines
- Ship Propulsion (Motors)
- SQUIDS (Superconducting Quantum Interference Devices)

Thomas Jefferson **Physics Olympiad**

A high school physics contest

TJ Physics Team <tjhsstphysicsteam@gmail.com>

The James Webb Space Telescope

If you have any questions or concerns, please email TJ Physics Team.

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Additional Assorted Problems	13
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Additional Resources	22

Rules

Teams will have from 9:00 PM EST on March 13th to 9:00 PM EST on March 15th to complete the TJPhO. All teams are required to submit their response with a cover page listing the team name and number, team member names, team email, and the date. Each submitted page should also have the problem number. All other formatting decisions are delegated to the teams themselves, with no one style favored over another. We suggest that teams use $\text{\LaTeX}$ for typesetting. The easiest way we've found to typeset $\text{\LaTeX}$ is with Overleaf.

Problems

Many of our problems will focus on the James Webb Space Telescope and the foundational physics that went into designing it. There is also a wild card section which has questions on an assortment of different topics – it is up to you to decide what approach you need to use. Lastly, we have a coding section, where in addition to doing some math you are expected to write a program in a language of your choice and submit your code for us to review.

Our problems were written by Tarushii Goel, Cyril Sharma, Anya Mischel, and Alvan Caleb Arlandu. We would like to thank Dr. Jonathan Osborne for reviewing our problems and providing valuable feedback.

Collaboration

Students participating in the competition may only correspond with other members of their team. No other correspondence is allowed, including mentors, teachers, professors, and other students. Teams are not allowed to use online/print resources or post content on online forums asking questions related to the exam. We welcome teams to email us if there are any questions or concerns. If a problem explicitly states that it is allowed, teams may use computational resources they might find helpful, such as Wolfram Alpha/Mathematica, Matlab, Excel, or programming languages.

Submission

Teams must submit their solutions by email to tjhsstphysicsteam@gmail.com by the deadline 9:00 PM EST on May 15th. Late submissions will not be considered. Solutions should be written in English and submitted as a single PDF document with the .pdf

extension. The email must contain “Submission” in the subject line. Only one person per team, the person who registered through the Google Form, should send the final submission.

Awards

The top five teams will receive awards.

1st place: \$200, 3 one-year subscriptions to WolframAlpha Notebook Edition

2nd place: \$150, 3 one-year subscriptions to WolframAlpha Notebook Edition

3rd place: \$100, 3 one-year subscriptions to WolframAlpha Notebook Edition

4/5th place: 3 one-year subscriptions to WolframAlpha Notebook Edition

Sponsors

We are incredibly grateful to our sponsors, Dr. Adam Smith and Mr. Robert Culbertson, and Thomas Jefferson High School for Science and Technology for their support. We would also like thank our sponsors WolframAlpha and the Thomas Jefferson Partnership Fund for making this Olympiad possible.

Introduction

The James Webb Space Telescope (JWST) was developed in a collaboration between NASA, the European Space Agency (ESA), and the Canadian Space Agency. It is named after James E. Webb, who was the administrator of NASA from 1961 to 1968. Development began in 1996 on a \$500 million budget with a launch initially planned for 2007. Instead it was launched last year and costed an esitimated total of \$9.7 billion.

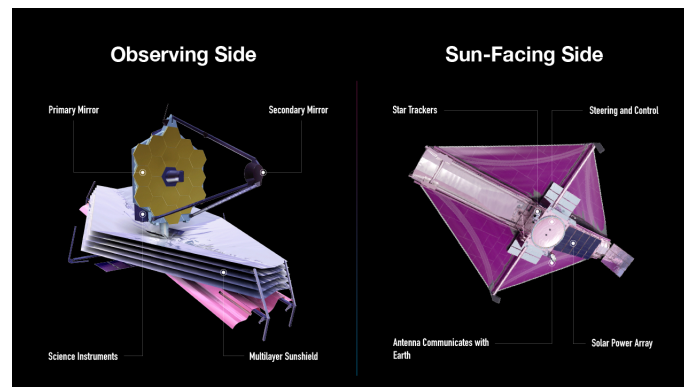


Figure 1: The James Webb Space Telescope

It is designed to perform near-infrared astronomy and is capable of seeing objects that are too faint and distant for the Hubble Space Telescope to capture images of. It can see as far back as 50 million years after the Big Bang.

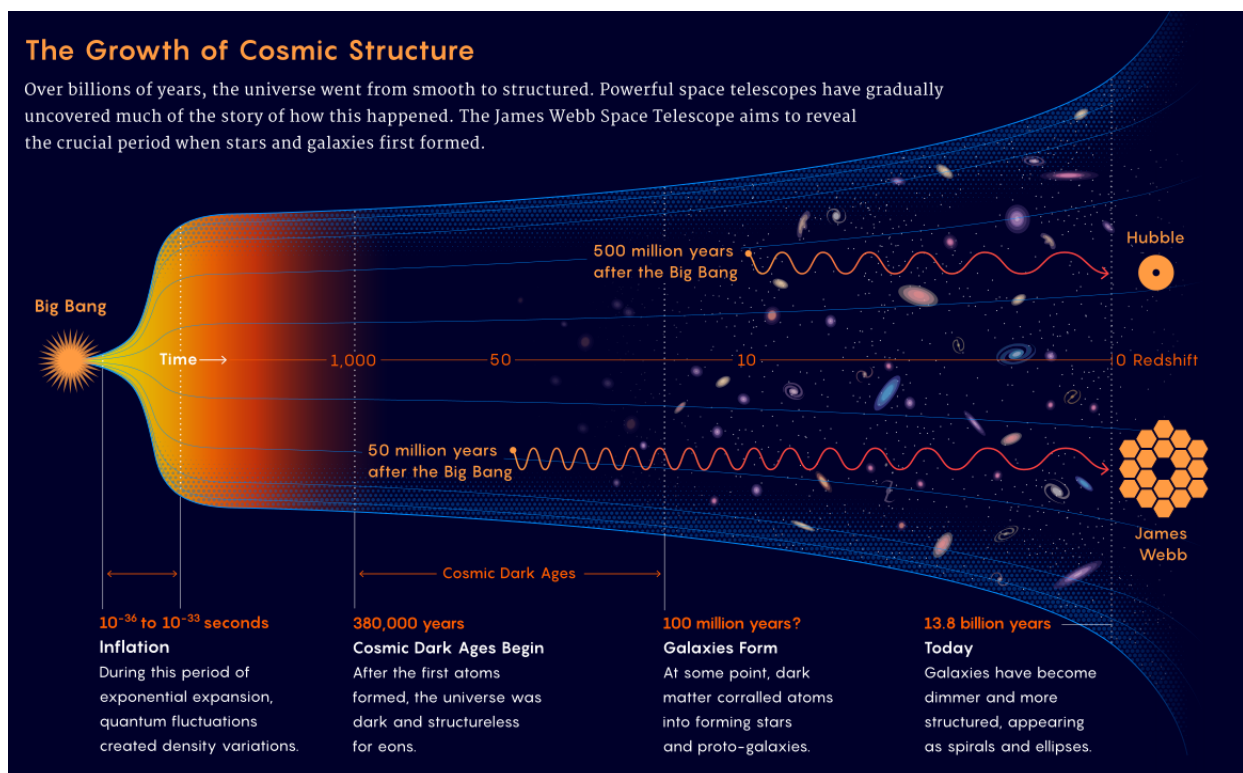


Figure 2: History of the Universe

The JWST was launched along with the Ariane 5 rocket on December 25th, 2021, from a ESA launch site in French Guiana. On December 27th, it sailed beyond the orbit of the moon, and on December 31st, it began the process of unfurling its massive sun shield and tightening the layers, a process that took until January 3rd, 2022. On January 24th, it finally arrived at the second lagrange point, L2, where it will remain while it takes high resolution images of space (previously, the Herschel Space Telescope and Planck Space Observatory have also orbited at the L2 point). The JWST has finished unfolding its mirrors and has spent the last few months aligning its optical instruments. Just a few days ago, it reached the end of its alignment process and took a high resolution image, which is included in Figure 4.

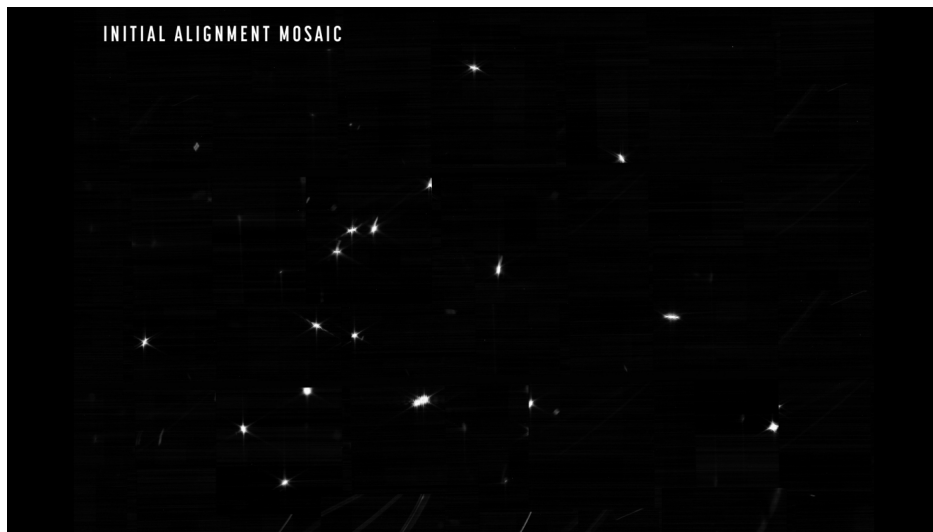


Figure 3: The first image taken by the JWST, at the beginning of the long process of alignment. The light is from a star named HD84406. Each of the 18 bright dots is the light reflected by one of the JWST's 18 primary mirror segments. (Image credit: NASA/ESA/CSA/STScI)

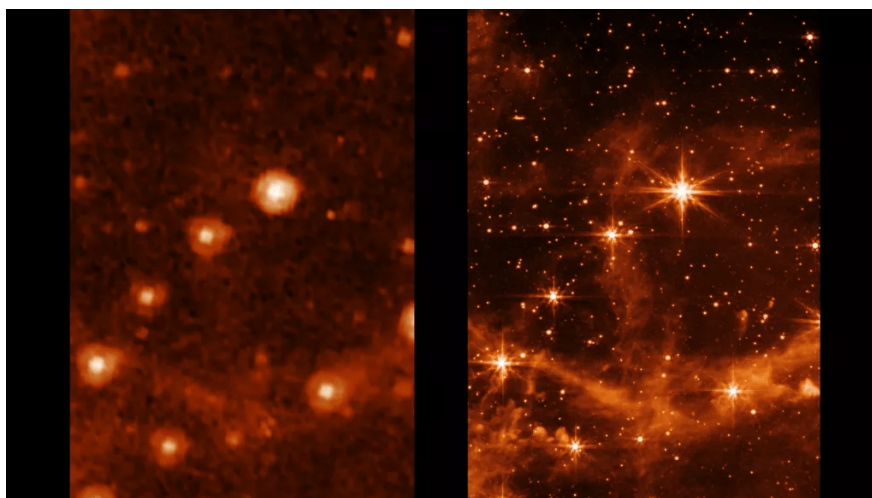


Figure 4: The Large Magellanic Cloud, as seen by NASA's Spitzer Space Telescope (left) and the JWST after a long period of alignment (right). (Image credit: NASA/JPL-Caltech (left), NASA/ESA/CSA/STScI (right))

The James Webb Space Telescope (JWST)

1. *Lagrange Points.* When communicating with a satellite, it's desirable if the satellite is always in the same place. Lagrange points are thus an attractive destination for satellites; as they remain stationary relative to the Earth (they orbit the sun with the same angular velocity as Earth). For this reason, the JWST has been sent to orbit the second Lagrange point, shown below.

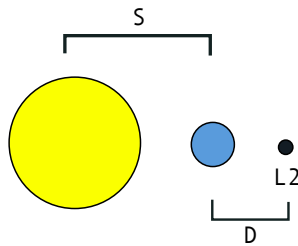


Figure 5: Lagrange Point

- (a) (1 point) Calculate the distance to L2, D , in terms of the mass of the earth and sun, M_e and M_s , and the distance of the sun from the earth, S . Hint: L2 does not move relative to the Earth, meaning it orbits the sun with the same angular velocity as the Earth.
- (b) (1 point) Calculate the gravitational potential energy as a function of the distance from the sun in the direction of the earth. You may assume the Earth and Sun are point masses. The JWST has mass m_j .
- (c) (1 point) The potential function calculated above doesn't account for the centripetal force needed to stay in the same relative position. Pick a frame of reference that rotates around the sun with the earth. From this frame of reference, a centrifugal force acts on all bodies away from the axis of rotation. Compute the effective potential function, a potential function that takes into account both the centrifugal force and the gravitational forces.
- (d) (1 point) Contrary to the calculated potential function, L2 is actually fairly easy to orbit. This is because the Coriolis force helps stabilize L2 orbits. The diagram below shows how the Coriolis force and gravitational forces balance, allowing an object to stay in a stable orbit around L2. Qualitatively explain how an object moved slightly towards the Earth from its spot on L2 would find itself at this steady state in terms of the Coriolis and gravitational forces.

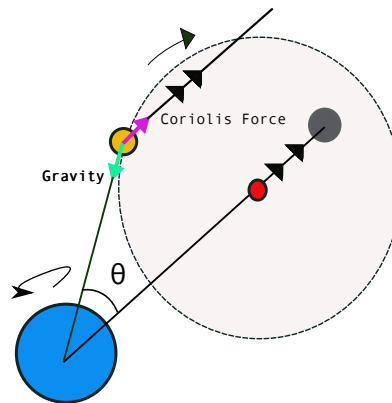


Figure 6: Stable Orbit of Lagrange Point

- (e) (1 point) Let the distance between the Earth and the Sun be S . Let the distance along the x axis from the Earth to the JWST be D . You may assume the sun's gravitational pull is not significantly angled, and is directed purely in x direction, as shown in the diagram. In terms of S and D , what angle θ must the JWST be at relative to the Earth in order to remain stationary relative to the Earth?

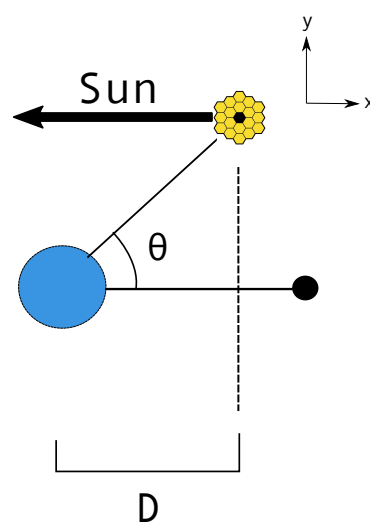


Figure 7

2. *Solar Radiation Effects.* The JWST's sensitive infrared imaging technology must be

kept at low temperatures to operate correctly. For this reason, the JWST has a sun shield made of Kapton E that blocks solar radiation. We can model the sun shield as a flat surface being struck by electromagnetic (EM) waves.

The energy flux of an EM wave is known as the Poynting vector, $\mathbf{S}$. This fluctuates with the frequency of the EM wave, but generally all we care about is the average of the magnitude of the Poynting vector, $\langle S \rangle$, which is also known as the light irradiance, I . The momentum density of EM waves is $\mathbf{P} = \frac{\mathbf{S}}{c^2}$. For this problem, assume that the Poynting vector is perpendicular to the shield.

- (a) (1 point) Express the average pressure p_a exerted on the sun shield as a function of I and c , assuming the shield absorbs 100% of the radiation. Then calculate the pressure p_r when it perfectly reflects the radiation.
- (b) (1 point) The shield is 22 by 10 m in size, its reflectance factor is 0.8, and the irradiance of light at the L_2 point, where the JWST is situated, is about 1366 W/m^2 (this is actually the irradiance of light at the Earth, but the L_2 point is only 0.01 AU away so the difference is negligible). Calculate the total force solar radiation exerts on the shield.
- (c) (2 points) Random fluctuations in solar irradiance can cause pressure gradients across the sun shield, which result in a torque. Use the set up shown below, where the force calculated in the previous question is directed towards the very end of the shield rather than evenly distributed, to calculate the torque on the shield. The effects of this torque need to be mitigated in order to keep the shield facing the sun and protect the JWST's sensitive equipment. One approach of doing so is to use a momentum wheel with stored angular momentum. Using the set up shown below, where the torque you calculated is exerted on the momentum wheel for a period of one day, to find the total rotation θ in terms of the angular momentum L of the momentum wheel, which is directed downwards on the figure below ($I_W \gg I_S$, where I_W and I_S are of moments of inertia about the COM of the wheel and shield for the wheel and shield, respectively). Graph θ vs. L . What is the axis of rotation?

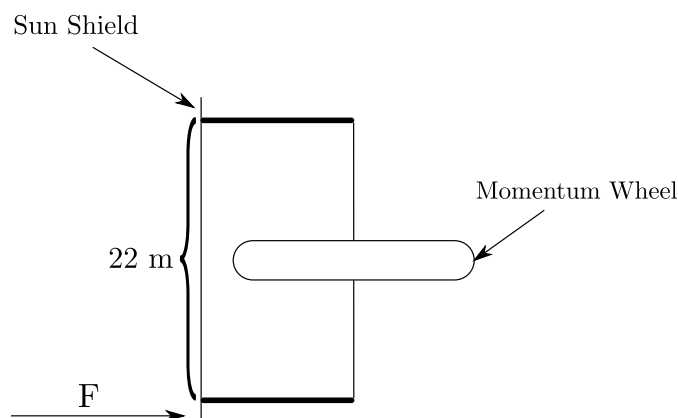


Figure 8: Top View of Shield-Wheel Configuration

3. (5 points) *Exoplanets*. Suppose the JWST discovers a faraway exoplanet orbiting a star in an elliptical orbit with eccentricity 0.42 and a semi-major axis of 1.9 AU. In order for this particular planet to have the possibility of carbon-based life, the

planet spend at least 20% of the time of its orbit within 1.4 AU of the star. Given this information, does the planet stay within the maximum habitability distance long enough to have the possibility to support life? *Wolfram Alpha or other integration-solving software is permitted for this problem only, but all steps including the integral expression must be explicitly written.*

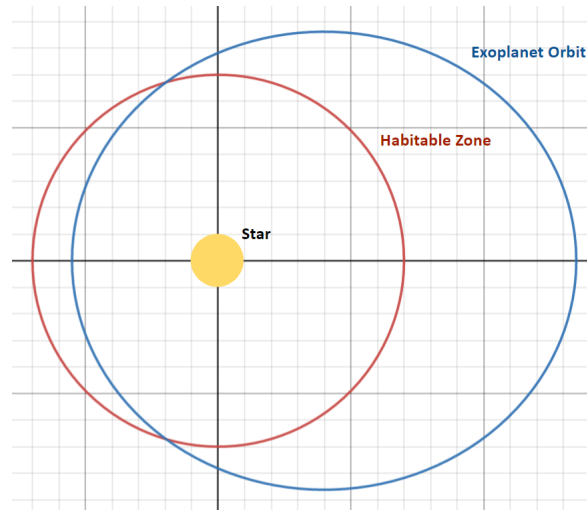


Figure 9: Exoplanet Orbit and Habitability Boundary

Imaging in the James Webb Space Telescope

4. *Lenses.* Lenses rely on the law of refraction, which states that light will bend when transitioning between two media. Specifically, $n_1 \sin \theta_1 = n_2 \sin \theta_2$, where n_1 and n_2 are the indices of refraction of the materials (phase speed $v = c/n$), and θ_1 and θ_2 are the angles that the incident and transmitted light rays make with the perpendicular to the boundary between the materials. JWST's lenses use refraction to concentrate light onto a focal point, thereby creating high resolution images.

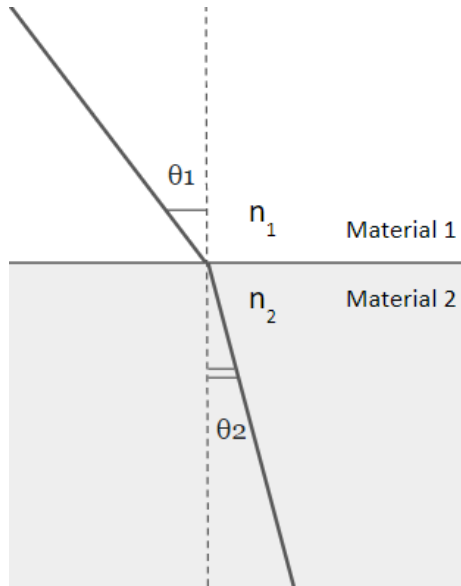


Figure 10: Refraction of Light Passing Through Different Materials

- (a) (3 points) Suppose you have a lens as shown in the setup below (gray region is glass), which focuses light from point source S in space at focal point P inside the lens. According to Fermat's principle of least time, both light rays shown must take the same amount of time to travel between the two points. By relating the travel times of the two rays, establish a relationship between s_1 and s_2 in terms of R , the radius of the lens, n_1 , and n_2 . Assume $d_1^2 - s_1^2 = (d_1 - s_1)2s_1$, $d_2^2 - s_2^2 = (d_2 - s_2)2s_2$, and $R^2 - x^2 = (R - x)2R$.

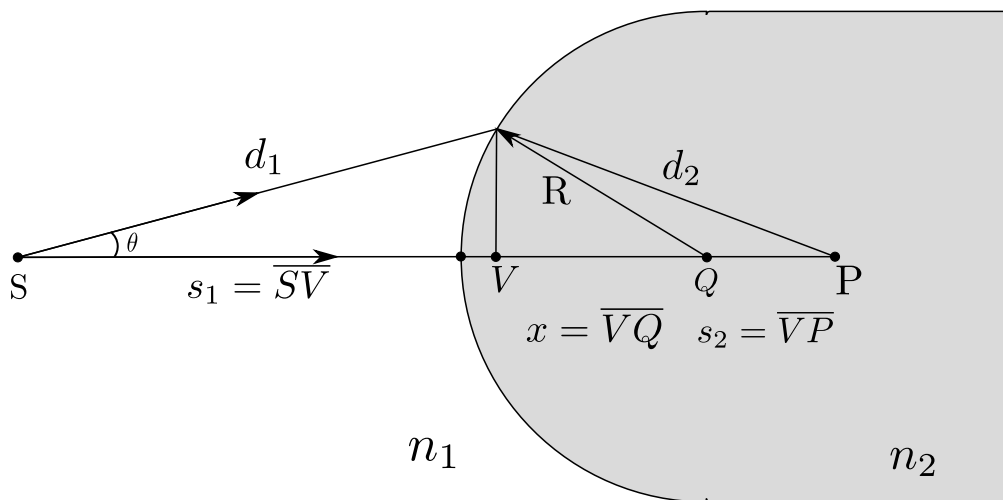


Figure 11: A Lens

- (b) (2 points) Replace the lens used in the previous question with a lens of the shape shown in the figure below, and derive a new expression relating s_1 and s_2 , in terms of R_1 , R_2 , n_1 , and n_2 . Assume $d \ll s_1, s_2$.

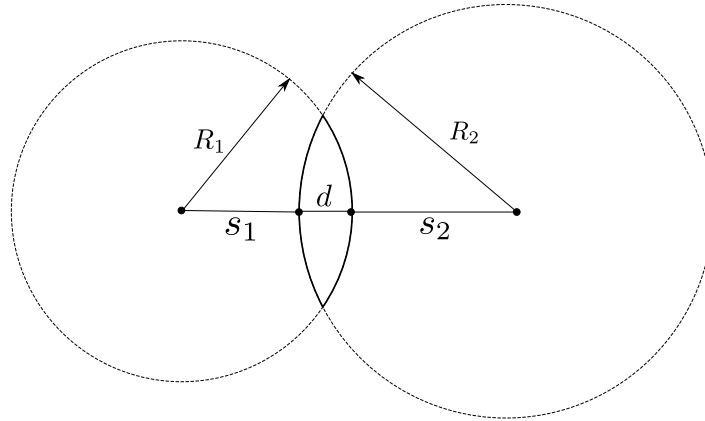


Figure 12: A Thin Lens

- (c) (1 point) Generally, the sources of light which a lens is engineered to focus are much farther away than from the lens than the focal points of those sources of light (e.g. the JWST images stars many light years away but the star image may be constructed a few centimeters away). The distance at which the image focuses is the f , the *focal length*, which is derived by letting $s_1 = \infty$. Plug this value for s_1 into your answer for the previous part to find the focal length.
- (d) (1 point) Light waves warp while traveling through space, and being able to measure these distortions and adapt optical systems accordingly is crucial. One method of measuring wavefront distortions is to arrange lenslets in a line, as shown in the figure below. Each lenslet will focus light on a different point on the image plane (see figure below), due to the difference in the angle at which the wavefront strikes the lens. Demonstrate how the lenslet array distinguishes between different wavefronts by marking the focal point of each lenslet for a planar wavefront. Then mark the focal points for the distorted wavefront, which may be shifted up or down. You should draw a total of 10 points, 5 on each image plane.

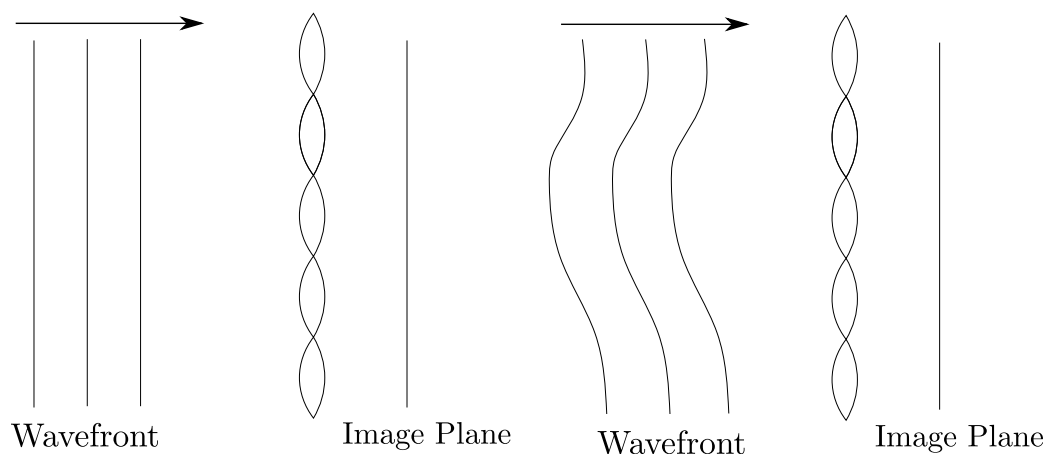


Figure 13: A Lenslet Array

5. *Diffraction.* The JWST's optical equipment is precisely fine-tuned so as to produce the highest resolution images possible. However, the resolution of these images has a theoretical limit due to the diffraction of light.

- (a) (3 points) Let's say you have a line of n harmonic electron oscillators spaced by a distance d that are moving in phase. The electric field created by single oscillator at a *large distance* r away from can be approximated as $A \cos\left(\omega t - \frac{2\pi r}{\lambda}\right)$, where λ is the wavelength of electromagnetic wave, and ω is its angular velocity. Write the resultant amplitude of the field oscillation A_R at a point a distance r away and at an angle θ (in terms of A , f , the frequency of oscillation, d , and θ). Note that in the diagram below the light rays reaching the point are shown as approximately parallel (and you can assume they are) because the point is very far away.

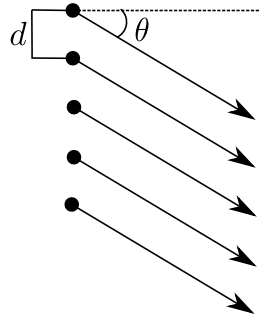


Figure 14: Electron Oscillator Setup where $n = 5$

- (b) (1 point) Plot I (the light irradiance, which is explained in the solar radiation problem) vs. θ . At what values of θ do you see the greatest irradiance of light? Where is the irradiance 0?
- (c) (1 point) Let's think of a one of the JWST's lenses as a small, circular opening in a barrier that allows light to pass through. The theory of diffraction states that the light passing through the opening is equivalent to that which would be generated by a lot of (really, infinitely many) sources of light of the same wavelength evenly separated by a small distance across the opening. So, the light that passes through the opening forms a diffraction pattern that is quite similar to that which you calculated in previous parts. Draw the diffraction pattern you would expect (it does not have to be mathematically precise, we are only looking for the general position of high irradiance and low irradiance regions).

Interestingly, the angle θ at which the first minimum in light irradiance occurs, measured from the direction of incoming light, is approximated by the formula $\sin \theta = 1.22 \frac{\lambda}{d}$ where d is the diameter of the aperture. So, using the Raleigh Criterion, θ is the smallest angular separation you can have between two objects while still being able to resolve them. This is what we mean when we say that the resolution of a camera is *diffraction-limited*.

Additional Assorted Problems

6. Windmills.

- (a) (1 point) Windmills generate energy by slowing down the wind and extracting its energy. Suppose we have a windmill as shown in the figure below. Assuming that we are dealing with incompressible, irrotational, and steady flow of air, which of the colored lines (red, blue, or black) best represents the general shape of the streamtube of air that passes through the windmill?

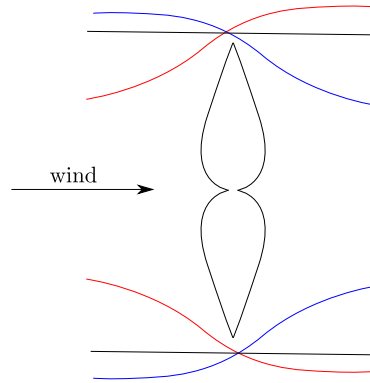


Figure 15: Top View of a Windmill

- (b) (2 points) Bernoulli's equation, which is derived from conservation of energy, states that $P + \frac{1}{2}\rho v^2 + \rho gh$ remains constant in steady flow. Assuming that energy is not inputted or outputted into the streamtube upstream or downstream of the windmill (only at the windmill), apply Bernoulli's equation to find the force exerted on the windmill. Express your answer in terms of W , the undisturbed wind speed, A_w , the area of the windmill, ρ , the density of air, and a , the induction factor, which is related to the design of the windmill and indicates much energy it extracts (specifically, the speed of air at the windmill is $W(1-a)$), and the unknown W_∞ , which is the speed of the air far downstream. Assume that gravitational potential energy is unchanging during fluid flow. Additionally, draw graphs of the pressure and wind speed vs. position along the streamtube (these only need to be correct to the first derivative).
- (c) (1 point) Derive another expression for the force on the windmill, this time applying conservation of momentum (in terms of the same variables as the previous part).
- (d) (1 point) Combine the expressions you found in the last two parts to find W_∞ in terms of W .
- (e) (1 point) Plug your expression for W_∞ back in to get a formulation for power extracted (assume 100% efficiency). What induction factor gives you the highest power? What is the ratio between the maximum power you can extract and the power of the wind passing through the windmill?
- (f) (1 point) Now let's consider a contraption composed of two propellers, one in the air and one in the water, connected by a shaft, as shown in the figure below. The analysis for the force on power extracted by the propeller in the air is exactly the same as that established in the previous parts. However, the propeller in the water must be reanalyzed (though the steps to do so are very similar) because it functions more like a fan than a windmill, speeding up the

water to a speed V at the propeller. Calculate the power delivered by the water propeller to the water in terms of ρ_2, A_2, V .

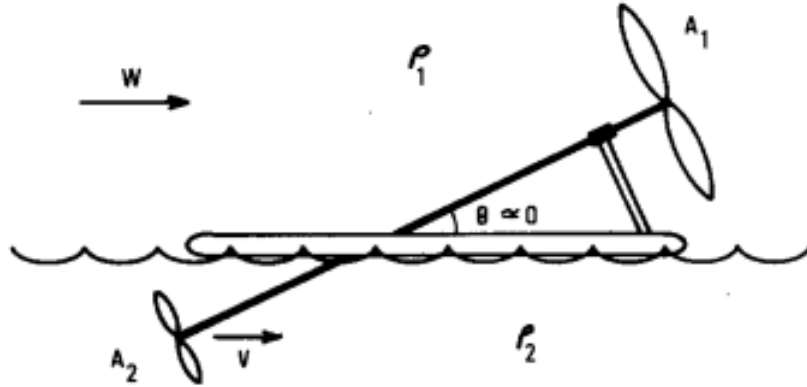


Figure 16: Propeller Contraption

- (g) (1 point) Assume $P_{in} = P_{out}$, and calculate the net force on the contraption using the induction factor that provides maximum efficiency (calculated in a previous part). Your answer should be in terms of a subset of these variables: $\rho_1, A_1, \rho_2, A_2, W$.
- (h) (2 points) What is the maximum speed of this contraption, in terms of W ? (the answer may be surprising)
7. *Ceaseless Cylinders*. Two infinitely long concentric cylindrical conducting shells are grounded (this means that they are connected to an infinite source/sink of charge, and charge can be induced on the conductors in order to maintain a potential of 0). The outermost conductor (conductor B) has a radius b , and the innermost conductor (conductor A) has a radius a . A single point charge q is placed at some radius r ($a < r < b$), inducing charge on the conductors (Q_A and Q_B).
- (a) (1 point) What is the total charge induced on both conductors, $Q_A + Q_B$? (in terms of some subset of the following variables: a, b, q, r). Provide a short explanation for how you can determine this without knowing Q_A or Q_B .
- (b) (6 points) Green's reciprocity theorem states that $\int_{allspace} \rho_1 V_2 dV = \int_{allspace} \rho_2 V_1 dV$, where ρ_1 and V_1 are the charge density and potential in one charge configuration, ρ_2 and V_2 are the charge density and potential in a completely separate charge configuration, and dV is the differential volume. Use this theorem to find Q_A and Q_B individually. (in terms of some subset of the following variables: a, b, q, r).

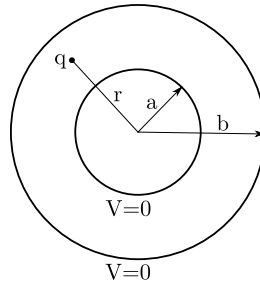


Figure 17: Sketch of the Situation

8. *Structures*. Stress represents the internal forces an object experiences, defined as the force over a cross-section divided by the area of that cross-section $\sigma = \frac{F}{A}$. Strain represents the elongation caused by a stress, defined as the change in length over the original length $\epsilon = \frac{\Delta l}{l}$.

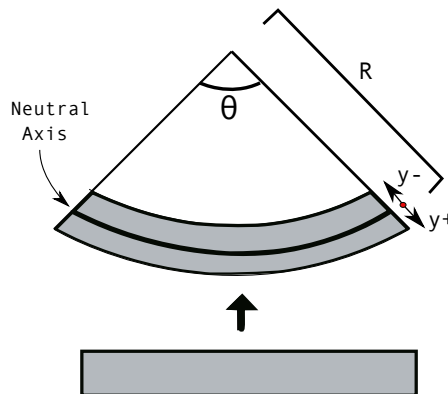


Figure 18: Bending Stress

- (1 point) In the diagram above, a beam is bent to have a radius of curvature R . Calculate the strain in the beam as a function of the displacement, y , from the neutral axis, where positive displacement takes you further from the center of curvature, as indicated in the diagram.
- (1 point) Stress is related to strain through Hooke's law, $\sigma = E\epsilon$, where E is Young's modulus and ϵ is the strain. Young's modulus is the property of the material the rod is made out of, and characterizes its response to stress. Using the above calculation for strain above, calculate the stress in the beam as a function of y , R , and E .
- (2 points) Using your expression for stress from part b, calculate the bending moment on the beam in terms of its moment of inertia I . Then plug your

answer back in to your calculation of stress to determine a formula for stress independent of the radius of curvature.

- (d) (2 points) The formula for stress you calculated above is used for calculating the bending stresses in a material. Another type of stress is known as direct stress, resulting from the application of a load (force over an area) which does not induce bending. In the diagram below, both bending stresses and direct stresses are present. Assuming the building has no strength under tension (an assumption that works well for structures made solely of concrete), what load displacement ϵ will cause the building to break under a load P ?

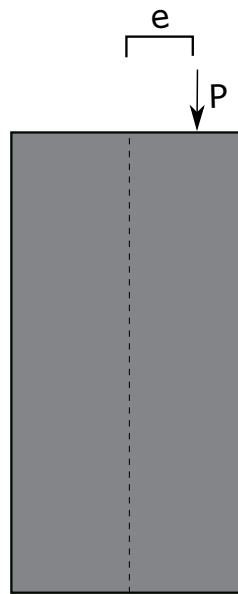


Figure 19: Building

9. *Intermediate Axis Theorem*

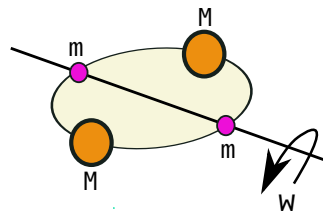


Figure 20

- (a) (1 point) In the above diagram, a thin, rigid, massless disk spins along the axis shown. The orange masses (M) have substantially more mass than the purple masses (m). When viewed from a reference frame that rotates with the disk (at angular speed ω), what centrifugal force per unit mass would a test mass m_t experience as a function of their distance (r) from the axis of rotation.

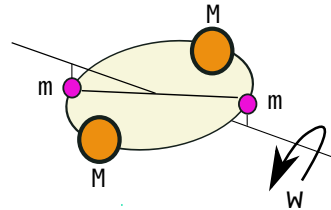


Figure 21

- (b) (3 points) Suppose the disk of radius R is disturbed an infinitesimal amount mid-rotation as shown in the above diagram. The purple masses now feel a centrifugal force. Calculate a differential equation which describes the angle of the small masses (m) with respect to the center of the disk as a function of time. Qualitatively describe the solution to this differential equation. You may assume the disk is in space, so no gravitational force, and that the tension forces are insignificant to the large masses, meaning the large masses stay in essentially the same location. Furthermore, since the purple masses are so small, you may assume the Coriolis force is not sufficient to overcome tension forces and thus can be ignored.

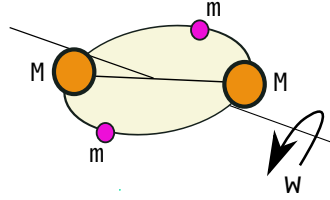


Figure 22

- (c) (1 point) The Coriolis force can be calculated as $2m(\vec{\omega} \times \vec{v}')$, where m is the mass of the object experiencing the force, $\vec{\omega}$ is the vector describing the angular rotation of the reference frame, and $\vec{v}'$ is the velocity of the object experiencing the force in the reference frame. Now suppose that the masses are disturbed as shown in the above diagram. Qualitatively explain how the Coriolis effect will modify the analysis from the previous part.

Coding Problems

10. *Grover's Algorithm.*

A qubit is the quantum equivalent of a classical bit. Like a classical bit, qubits can only be measured in one of two states, denoted 0 and 1. Unlike a classical bit, the state of a qubit is not determined before the measurement. Before measurement, qubits exist in a state that is a superposition of the measurable states, which after a measurement, collapses problematically into one of the two measurable states.

A qubit's state is a vector of two complex numbers $\begin{bmatrix} a + bi \\ c + di \end{bmatrix}$. Each number in the state vector is associated with a specific state. Typically the first number is associated with $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and the second is associated with $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The probability of a qubit collapsing into a state is given by the magnitude squared of the associated component in the state, for example, if my state vector is $\begin{bmatrix} a + bi \\ c + di \end{bmatrix}$, then to calculate the chance the qubit will collapse into the state $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is $|a + bi|^2 = (\sqrt{a^2 + b^2})^2 = a^2 + b^2$. Since the qubit must collapse into one of the measurable states, the sum of the magnitudes squared (the sum of the probabilities) for all the qubit's state must add up to 1.

More generally, a quantum system can be represented as a vector of amplitudes that represent the probability of collapsing into a given state. For example, a quantum

system of two qubits can be measured to be in one of four possible states: $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix},$

$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$ A possible representation for the state could be $\begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}.$ The magnitude

squared denotes the probability of being in a given state, so the probability of being in

state $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ is $|0.5|^2 = 0.25.$ Note that the sum of the probabilities add to 1, $0.25 * 4 = 1,$

as the qubit must collapse into some state.

Furthermore, if vectors are representation of quantum states, matrices are representations of quantum transformations, processes that convert one quantum state into another.

- (a) (1 point) Suppose we have a quantum system with 2^N possible states. Let the initial state of the system be s the equal superposition state s . Consider the pure target state w and the equal superposition state s' that is orthogonal to w . If $s = c_1 w + c_2 s'$, find c_1 and c_2 .
- (b) (2 points) U_s , also known as the diffuser, is a transformation computed through the formula $U_s = 2ss^T - I$, where s is the initial state and s^T is the transpose of s . Compute $U_s s$. Compare the effect of U_s on s with U_s on w where w is orthogonal to s .
- (c) (1 point) Let U_f be the transformation which negates the amplitude of the target state. Compute $U_s U_f s$.
- (d) (1 point) Each application of $U_f U_s$ induces a rotation on the input state, calculated above as θ . For what n is $\|(U_f U_s)^n s - w\|$ minimized? In other words, how many applications of $U_f U_s$ are necessary to rotate the input state, s as close to the target state, w as possible? Show your derivation.
- (e) (3 points) **Code:** The procedure outlined in the previous steps, applying U_f and U_s repeatedly to a state are known as Grover's algorithm. Write a computer program that utilizes Grover's algorithm to find a provided target state, say $|100010\rangle$ given an arbitrary number of starting states. Specifically, write a function that takes in a target state and a value representing the total number of states. Your function should represent the operators U_s and U_f as matrices, and should output the probabilities of measuring the target state after the optimal number of iterations and the number of iterations your algorithm used.

11. JWST Collision

Suppose when the Earth is at the point closest to the Sun, the perihelion, the JWST, with a mass of 6161.4 kg, is stationed at the L2 point in stable orbit. Suppose a rogue satellite, of mass 2547.3 kg, collides with the JWST such that both satellites remain intact. At the point of instantaneous collision, the velocity of the asteroid is $3000m/s$, parallel to the velocity of the JWST which we consider to be in the $+\hat{j}$

direction. For the below problems, use the standard numerical values for the mass of the Earth and Sun.

- (a) (1 point) If the collision is perfectly inelastic and the asteroid's velocity is in the opposite direction of the JWST's, compute the velocity vectors of the JWST and the rogue asteroid immediately after the collision.
- (b) (1 point) If the collision is elastic and the asteroid's velocity is in the same direction of the JWST's, compute the velocity vectors of the JWST and the rogue asteroid immediately after the collision.
- (c) (2 points) The coefficient of restitution is defined as the ratio of the final to initial relative speeds between the two objects. If two objects with velocities u_a , u_b and masses m_a , m_b respectively collide with coefficient of restitution C_R , derive the final velocities v_a , v_b of both objects after impact.
- (d) (4 points) **Code:** Assuming elastic collision, plot the trajectory of both the JWST and the rogue satellite through the first 30 days after collision. Provide the final positions of both satellites at the end of the time period. Explain with reasoning what type of equilibrium exists at the L2 point?
- (e) (2 points) **Code:** Assuming elastic collision, Plot the distance of the JWST to the L2 point over the first year after collision. Provide the final position of the JWST at the end of the time period.

Additional Resources

Below is a list of additional information about the James Webb Space Telescope and its recent advances for those interested. Click on the hyperlink to visit the page.

Design of the JWST

- [Optics on the JWST](#)
- [Full Summary and Detailed Design Overview](#)
- [Orbit of the JWST](#)
- [Operating Temperature](#)

JWST Timeline

- [Timeline of JWST](#)

Thomas Jefferson **Physics Olympiad**

A high school physics contest

This exam covers concepts related to chaos and deviations from equilibria and explores current insights into this topic and applications.

TJ Physics Team <tjhsstphysicsteam@gmail.com>

Deviations from Equilibria

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Rules

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The winning teams will be decided by the number of points. If two teams have the same number of points, the higher-ranking one will be the one who submitted first.

Even if you are not able to fully solve a problem, please write up what you can. We want to see how you approached the problem, and partial credit may be awarded.

Collaboration

Students participating in the competition may only correspond with other members of their team. No other correspondence is allowed, including mentors, teachers, professors, and other students. Teams may use any computational resources they might find helpful, such as Wolfram Alpha/Mathematica, Matlab, Excel, or programming languages (C++, Java, Python, etc). Teams are not allowed to use online/print resources with the exception of language documentation for any problems that may require computation. Teams are under no circumstances allowed to post content on online forums asking questions related to the exam. We welcome teams to email us if there are any questions or concerns.

Submission

Teams must submit their solutions through the provided Google Form by the deadline 11:59 PM EST on May 30th. Late submissions will not be considered. Solutions should be written in English and submitted as PDF documents with the .pdf extension. Only one person per team, the person who registered through the Google Form, should send the final submission.

Awards

The top five teams will receive 3 one-year subscriptions to WolframAlpha Notebook Edition

Sponsors

We are incredibly grateful to our sponsors, Dr. Adam Smith and Dr Abdel Hady Ebrahim, and Thomas Jefferson High School for Science and Technology for their support. We would also like thank our sponsor WolframAlpha for making this Olympiad possible.



Introduction

One fascinating aspect of our world is that many natural phenomena follow cyclical rules and regulations. For instance, a physics student will likely be familiar with simple harmonic motion: the movement of a simple pendulum, a simple harmonic oscillator, can be defined by a differential equation, and its period can be calculated. What happens, though, when these cycles are disturbed? Additionally, in what situations do these cycles appear?

Having been studied for centuries, equilibria appear in many disciplines, including but not limited to oscillations, hydrostatics, topography, astrophysics, and thermodynamics. Furthermore, there are many modern topics built off of equilibria, such as chaos theory and statistical physics. Current popular and intriguing questions in physics that pertain to equilibria include the notable Three Body Problem and Double Pendulum Problem.

The TJ Physics Team is proud to announce that the fifth annual TJ Physics Olympiad (TJPhO) focuses on this very topic! This year's contest spans a wide range of subjects, from the well-known pendulums to the lesser-known hydrostatic equilibria. We hope that our examination can introduce you to new physics topics in this fascinating world of deviations from equilibria, as well as provide a fun learning experience.

Happy exploring!

TJ Physics Team Officers

1 Problems

Hydrostatic Equilibrium (10 pts.)

1

Hydrostatics is the study of fluids at rest. The concept of hydrostatic equilibrium and its application in fluids and astrophysics will be explored in this problem.

- a) (1 point) The Hydrostatic Equation, $\frac{dp}{dz} = -\rho g$ states that the pressure in a fluid at hydrostatic equilibrium varies constantly in respect to change in height. Given a uniform box of air in the Earth's atmosphere with density ρ , base area A , and height dz^* , derive the Hydrostatic Equation.
- b) (4 points) Hydrostatic equilibrium also applies to stars. Given a star of mass M , radius r , and density function $\rho(r)$, derive a Hydrostatic Equation similar to the one in (a) regarding how the pressure in a star varies in respect to the distance to its center.
- c) (3 points) The earth's density is not constant, but varies approximately according to $\rho(r) = C \left(1 - \frac{r}{R}\right)$, where C is a constant, R is the radius of the Earth, and r is the distance from the Earth's center. Using your equation from (b), find pressure as a function of the distance from the Earth's center.
- d) (2 points) If the pressure of Earth's inner core is increased by an amount ΔP , find Earth's new gravitational field in terms of g .

Tunnel Through a Planet (7 pts.)

2

One popular thought experiment is, what if a tunnel was dug through the middle of the Earth? For this problem, consider a tunnel that was dug through the middle of a fictional Planet Epsilon. The density of the Planet Epsilon is constant at 3 kg/m^3 . Approximate the radius of the Planet Epsilon as 12000 km.

- a) (3 points) Assuming negligible friction, if a person modeled as a point with mass 50 kg fell into the tunnel from the surface of Planet Epsilon, what would be their period of oscillation?
- b) (4 points) A person of mass m is placed, stationary, in this tunnel a distance of x away from the center of Planet Epsilon. The person is given a slight push toward the center. Find the person's position as a function of time.

Pendulum with Drag Force (8 pts.)

3

Let there be a pendulum set up with a massless string of length 1 meter and a mass of 1 kg hanging.

- a) (2 point) The pendulum is displaced by an angle of 90° . What is the maximum speed of the pendulum? Ignore air resistance.
- b) (3 points) Now let us take into account air resistance. Let the force of air resistance be given by $F = \beta v^2$ with $\beta = 0.01$ kg/m. What is the maximum speed of the pendulum? Feel free to use computer software.
- c) (3 points) Eventually, the pendulum will come to a near rest. At what time will the pendulum have lost 99% of its energy?

Hookean Springs (10 pts.)

4

Consider a spring with a linear restoring force of the form $F = -kx$, where k is a constant and x is the displacement from the equilibrium point. Suppose a block of mass m is attached to the end of the spring. The block is pulled a distance A from its equilibrium point and released.

- a) (2 point) Using Newton's Laws, set up a differential equation that relates the position x to the amount of time t that has passed after the block is released.
- b) (2 point) Assume that the solution for x has the form $x(t) = A \sin(\omega t)$, for some constant ω . Using this assumption, determine the value of ω and the period of oscillation of the block.

Now we consider the same problem as in part a, but we now take a slightly different approach. This new method will be useful later on.

- c) (2 point) Let E be the total energy of the system. When the block is stretched at the maximum displacement A , we have $E = \frac{1}{2}kA^2$. Consider the block at a general position x . Use conservation of energy to set up a differential equation using the variables from part a.
- d) (4 points) Using your equation from the previous subpart, find the period of oscillations of the block (note that to compute the integral through the differential equation, you may have to use a trigonometric substitution).

Nonlinear Spring Forces (10 pts.)

5

We now look into springs that do not necessarily obey Hooke's Law. Consider a spring with a nonlinear restoration force of the form $F = -kx^n$. Again, suppose a block of mass m is attached to the end of the spring and is pulled a distance A from its equilibrium point.

- a) (2 points) When the block is released, the block will exhibit harmonic motion. Using dimensional analysis, determine how T varies with A in terms of n .
- b) (2 points) When the block is pulled a distance A from its equilibrium point, the period of oscillations is T . Suppose the block is pulled an extra distance ΔA , with $\Delta A \ll A$. As a result, the period of oscillations changes by ΔT . Find ΔT in terms of ΔA , A , T , and n .
- c) (3 points) Using the same procedure as in problem 4 parts c and d, use conservation of energy to set up an integral expression for the period of the subsequent oscillation. Do not evaluate the integral.
- d) (3 points) Considering the special case of $n = 2$, use a calculator to find the period of the oscillation. Write your answer in the form $T = cf(m, k, A)$, where f is a function solely dependent on m , k , and A , and c is a constant expressed to three significant figures.

Pendulums (11 pts.)

6

Consider a bob of mass m attached to one end of a string of length L . The other end of the string is fixed to the ceiling. The bob is pulled from its equilibrium position by an angle θ and released.

- a) (1 point) Suppose $\theta \ll 1$. Find the period of oscillations of the subsequent motion.

Now assume that θ is not negligible, so we cannot use small angle approximations.

- b) (3 points) Using conservation of energy, determine an integral expression for the period of oscillations.

The earth is a rotating reference frame. As a result, a pendulum may experience deflection forces, most notably from the Coriolis force. Suppose the earth rotates at angular speed Ω .

- c) (1 point) Suppose the pendulum is placed at the equator of the earth. Find the period of oscillations of the subsequent motion. You may use small angle approximations.

Now assume that the pendulum is moved to the North Pole. The pendulum is initially at rest at the origin $(0,0)$. Now the pendulum bob is pulled from its equilibrium position by a small angle θ , with $\theta \ll 1$.

- d) (2 point) At a general coordinate position (x, y) , determine the Coriolis forces in the x and y directions. Be sure to clearly indicate the direction by including a negative sign when appropriate.
- e) (2 point) The tension in the string of the pendulum also provides a restoring force. Find approximate expressions for the restoring tension forces in the x and y directions.
- f) (2 points) Combining your results from the previous two subparts, determine a system of two differential equations representing the motion of the pendulum.

Classifying Equilibria (10 pts.)

7

If an object in a system is in equilibrium, that equilibrium can be classified into three main categories: stable, unstable, and neutral. If an object is in stable equilibrium, any small perturbations to the system result in forces that restore the object to its equilibrium point. If an object is in unstable equilibrium, any small perturbations to the system result in forces that cause the object to move even farther away from its equilibrium point. If an object is in neutral equilibrium, any small perturbations to the system result in the object staying stationary in its newly displaced position.

An object is in equilibrium if the net force acting on that object is 0 (assuming no torques). For each of the three given forces F as functions of positions x given below, determine all equilibrium points, and classify them as either stable, unstable, or neutral.

- a) (1 point) $F(x) = x(x-1)(x-2)(x-3)(x-4)$
- b) (1 point) $F(x) = \sin(x^2)$
- c) (2 points) $F(x) = \sin(x^3)$

Equilibrium points, as well as their classifications, can also be found through potential energy functions of the system. For each of the three given potential energy functions (potential energy U as functions of position x) below, determine all equilibrium points, and classify them as either stable, unstable, or neutral.

- d) (1 point) $U(x) = x^3 - x^2 - x$
- e) (2 points) $U(x) = e^x \sin(x)$
- f) (3 points) $U(x) = C((k/x)^{12} - (k/x)^6)$, where C and k are constants. This potential energy function is known as the Lennard-Jones model, which gives the potential energy between two atoms in a diatomic molecule.

Double Pendulum Pandemonium! (30 pts.)

8

You've probably seen a single pendulum, but when we add a second, things start to get wacky. Consider a double pendulum hanging from a ceiling.

- a) (1 points) Draw an FBD and parameterize each mass with respect to the angle from each pivot.
- b) (4 points) Give any closed-form differential equation for the angular position that could, hypothetically, be solved.
- c) (2 point) Use Euler's integration to solve for $t = 10$ seconds given $m_1 = 1.5$ kg, $m_2 = 2$ kg, $\theta_1 = 30^\circ$, and $\theta_2 = 45^\circ$. What is the bound for the expected error?
- d) (2 points) Google Runge-Kutta 4/5. Runge Kutta 4/5 is a 4th order numerical integration. For this problem alone you may reference Python or other language documentation and the Wikipedia reference on the topic. Implement the method and solve. What is the bound for the expected error?
- e) (5 points) Create a simulation of a double pendulum in the framework of your choice. Submit an animation or website of it working.
- f) (1 points) Count the number of times the pendulum flips. Plot this number using colors on a grid of varying θ from 0 to $\frac{\pi}{2}$. Do you notice any interesting patterns?

Now consider an inverted double pendulum.

- g) (15 points) Repeat all the steps above for this type of pendulum.

2 Further Resources

For those interested in further resources, here is an enumeration of articles and web pages regarding current equilibria applications and news, as well as resources for learning about equilibria. Click on the hyperlinks for more information.

- [Equilibrium Statistics Models](#)
- [Current Flow While in Equilibrium](#)
- [MIT OpenCourseWare Double Pendulum Lecture](#)
- [Society for Chaos Theory Resources](#)
- [Dwarf Planets and Hydrostatic Equilibrium](#)
- [Thermochemical Equilibrium](#)
- [The Three Body Problem](#)
- [Three Body System: Experimental Electronic Realization](#)
- [Chaos Theory Overview](#)

Thomas Jefferson **Physics Olympiad**

A high school physics contest

This exam covers concepts related to astrophysics, specifically stars, and explores how physics can be used to analyze the properties of these celestial bodies.

TJ Physics Team <tjhsstphysicsteam@gmail.com>

Stars and Astrophysics

If you have any questions or concerns, please email TJ Physics Team.

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Physics with Elliot.

Sponsors

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Introduction

In the few hundred million years following the big bang, stars began to form as clouds of hydrogen collapsed due to gravity. In the cores of these stars, fusion processes began to produce increasingly heavy elements, while releasing light and heat into the rest of the universe. When stars reached the end of their life cycles, they shed stellar material in the form of planetary nebula or supernovae, which allowed future stars to form. Over billions of years, generations of stars formed and died out. In that process, planets formed in orbits around these stars, and in the case of the Earth, were able to support life. Throughout human history, people have looked up into a night sky filled with stars. During the last century, advances in science and technology have allowed humanity to gain an improved understanding of these celestial bodies. From our familiar sun, to extreme objects such as black holes where our understanding of physics break down, stars play a pivotal role in the universe.

The TJ Physics Team is proud to announce the sixth annual TJ Physics Olympiad (TJPhO)! We hope this contest will introduce you to the fascinating field of stars and astrophysics. To review concepts commonly used in astronomy to observe stars, we have included review problems covering topics such as stellar parallax, radiation laws, and nuclear fusion processes.

This contest covers a wide range in stellar physics including radiation pressure, white dwarf stars, binary star systems, hydrostatic equilibrium, and supernova explosions. We hope you have as much fun learning about stars as we had writing this contest.

Happy exploring!

TJ Physics Team Officers

1 Background Problems

Gravitation (40 pts.)

1

There always exists an attractive gravitational force between any two bodies with mass. *Newton's Law of Universal Gravitation* states that

$$\mathbf{F} = -\frac{Gm_1m_2}{|\mathbf{r}|^2} \hat{\mathbf{r}},$$

where $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$ is the gravitational constant, m_1 and m_2 are the masses of the two bodies, and $\mathbf{r}$ is the displacement vector between the two bodies. The negative sign simply indicates that the gravitational force is always attractive.

Gravity is a conservative force (work done is independent of path), meaning that it has an associated potential. The *gravitational potential energy* of a two-object system is given by

$$U = -\frac{Gm_1m_2}{r},$$

where r is the distance between the two objects.

Suppose an object of mass m orbits around another object of mass M . The *total energy* E is given by

$$E = -\frac{GMm}{2a}$$

for circular and elliptical orbits, where a is the semi-major axis of the orbit.

For all parts to this question, express your answer in terms of m , M , r , R , and any fundamental constants. Remember that you may use integral calculators as needed.

- A spacecraft of mass m orbits around a stationary star of mass M and radius R such that $m \ll M$. The orbit is completely circular with a radius of r . Find the velocity of this spacecraft.
- We say that the spacecraft “falls into the star” when it’s distance from the center of star is less than R ($r < R$). What is the maximum angular momentum for this to occur at some point in the spacecraft’s orbit, assuming the maximum distance from the star is still r ?
- If the spacecraft’s initial orbit was circular with radius r , what instantaneous change in the velocity would it need to undergo to achieve the orbit from part b?
- Find the period of the orbit in part b), where the period is the amount of time it takes the spacecraft to complete one orbit.

- e) Now suppose that at time $t = 0$, an identical spacecraft located a distance r from the center of the star has no initial velocity. At what time t would the spacecraft reach the surface of the star?

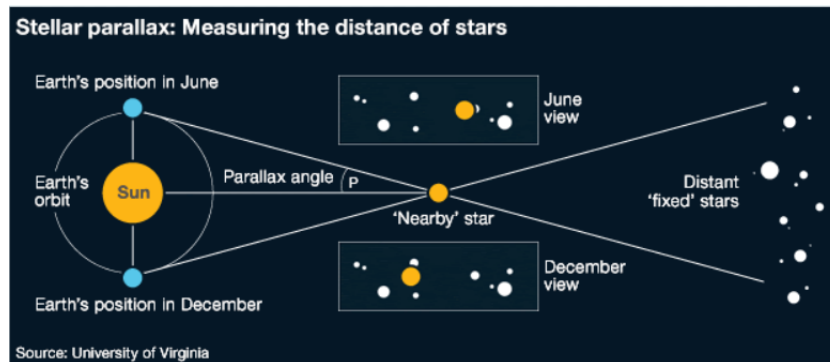
Parallax and Intensity (30 pts.)

2

In the early 19th century, it was believed that stars reflected the sun's light instead of producing their own. For this to be possible, stars would have to be much closer to the sun than they actually are. In this problem we will investigate how the intensity of light changes with distance and how to calculate distances to nearby stars.

As the Earth revolves around the sun, the positions of nearby stars in the night sky shift slightly compared to very distant stars (which are considered to be completely stationary). Knowing the radius of the Earth's orbit around the sun, and assuming the orbit is perfectly circular, one can calculate the distances to nearby stars.

The stellar parallax angle is the angle representing the shift in the star's position. The maximum angular change in the star's position is twice the parallax.



The intensity of the star is defined as the rate of energy transfer per area:

$$I = \frac{P}{4\pi d^2}$$

where I is the intensity of the starlight, P is the power emitted by the star, and d is the distance from the light source.

- Tau Ceti is a star with an observed parallax of 0.2738 arc seconds. An arcsecond is equal to $1/3600$ of a degree. The radius of the Earth's orbit is 1.496×10^{11} m. What is the distance of Tau Ceti from the Earth?
- The intensity of the light from Tau Ceti is 1.176×10^{-9} W/m². Find the power outputted by that star.

- c) The power outputted by the sun is on the order of 10^{26} W. The radius of Tau Ceti is 5.5×10^8 m. If Tau Ceti reflected all of the light hitting its surface, estimate to an order of magnitude the intensity of the reflected light reaching Earth from Tau Ceti.

The minimum intensity of light that humans can see is around 10^{-10} W/m². If stars only reflected the sun's light, they would appear far too dim for humans to see. The discovery of parallax provided the first evidence that stars produced their own light.

Radiation Laws (30 pts.)

3

Stars can be approximated as ideal blackbodies, which means that they follow Wein's Displacement Law and the Stefan-Boltzmann law.

All ideal blackbodies radiate some radiation in all wavelengths, but the vast majority of this radiation is close to a certain wavelength. The peak wavelength of the light is the one with the highest intensity. Wein's Displacement law relates the temperature of a star and its peak wavelength by:

$$\lambda T = b$$

where b is a constant equal to 2.898×10^{-3} m · K.

The Stefan-Boltzmann law is a statement on the power radiated by an ideal blackbody:

$$P = \sigma AT^4$$

where σ is a constant equal to $5.67 \times 10^{-8} \frac{\text{W}}{\text{K}^4 \cdot \text{m}^2}$ (Watts per kelvin to the fourth per meters squared).

- a) Arcturus is a red-orange star 36.7 light years from earth (one light year is 9.46×10^{15} m) with peak wavelength of 676 nm and intensity of light of 4.33×10^{-8} W/m². Calculate the radius of the star.

Stars like Arcturus are known as red giants due to their large size. At the end of the sun's life, its outer layers will expand to past the orbits of Mercury, Venus, and possibly Earth. Even larger are red supergiants. One example of such a star is Betelgeuse, with a radius of 617 million kilometers. If placed at the center of our solar system, Betelgeuse would engulf all four terrestrial planets, and its surface would nearly engulf Jupiter.

- b) The sun has a power output of 3.85×10^{26} W and is 1.51×10^{11} m away from Earth. Neglecting the greenhouse effect, if the Earth did not reflect light, what would be its equilibrium temperature? You may assume that the far side of the Earth has the same temperature as the near side.

- c) In reality, the Earth has an albedo of 0.3, which means that 30% of the light reaching the Earth is immediately reflected. Find the equilibrium temperature taking into account effects from the albedo.

The Earth remains at a habitable temperature thanks to the greenhouse gases in the atmosphere, which absorb and re-emit radiation that was reflected and radiated from the surface. Through this “trapping of heat” the Earth’s equilibrium temperature is significantly higher than it otherwise would be.

Nuclear Fusion (40 pts.)

4

Stars must generate vast amounts of power to emit as radiation. This energy is supplied inside the stellar core, in which temperatures are so high that nuclei can undergo nuclear fusion. During this process, two smaller nuclei are smashed together to form one larger nuclei, which has a mass slightly less than the sums of the two original masses. The rest energy of a particle is $E_0 = mc^2$, so this change in mass is converted to energy that is released by this reaction.

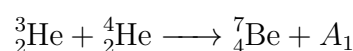
Nuclear fusion involves subatomic particles, which primarily include leptons and baryons. The leptons relevant to nuclear fusion in stars are electrons (denoted as e^-), electron neutrinos (ν_e), and their anti-leptons (e^+ , $\bar{\nu}_e$). The leptons have a lepton number of 1 and the anti-leptons have a lepton number of -1 . Electrons have a charge of -1 , neutrinos have a charge of 0, and their anti-particles have the opposite charge.

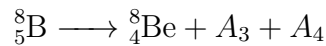
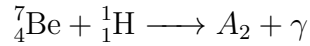
Baryons include protons, which have $+1$ charge, and neutrons, which have no charge. Protons and neutrons have baryon numbers of 1 while their antiparticles have baryon numbers of -1 .

In a reaction involving these subatomic particles, charge, lepton numbers, and baryon number must be conserved.

A nuclear reaction can release energy in the kinetic energy of these particles, or in the form of radiation (denoted as γ) carried away by massless particles called photons.

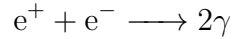
- a) The proton-proton chain (also known as the pp chain) is one of the reactions in which protons undergo nuclear fusion, eventually resulting in alpha particles (helium nuclei). The pp-III chain is one mechanism by which this could occur. Although it does not contribute to a large portion of the sun’s energy output, the pp-III chain produces highly energetic neutrinos which scientists can more easily detect. The pp-III branch consists of the reactions with missing particles A_1 , A_2 , A_3 , and A_4 (A_3 and A_4 are interchangeable):





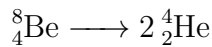
What are the particles/nuclei represented by A_1 , A_2 , A_3 , and A_4 ?

- b) (2 point) Positron electron annihilation accounts for some of the released energy. The equation for this process is:



The mass of electrons and positrons are 9.11×10^{-31} kg. What is the energy released in this process?

- c) (2 point) ${}^8_4\text{Be}$ quickly decays into two two alpha particles:



The atomic masses of ${}^4_2\text{He}$, ${}^3_2\text{He}$, and ${}^1_1\text{H}$ are 4.0015 amu, 3.0160 amu, and 1.0073 amu, respectively. One amu is equal to 1.6605×10^{-27} kg. What is the energy released from one nucleus of Helium-3, an alpha particle, and a proton undergoing this process?

- d) Neutrinos can be detected in large observatories, where the neutrinos excite a charge lepton in a large pool of water. The charged lepton then moves faster than the speed of light in the medium, which produces Cherenkov radiation. By analyzing the intensity and angle of the Cherenkov light, scientists can deduce the speed of the particle in the medium. A neutrino has a velocity related by $\frac{v}{c} = 1 - 3.64 \times 10^{-5}$. The neutrino mass is $0.120 \text{ eV}/c^2$. What is the total energy of the neutrino?

The pp-III and pp-IV cycles can produce neutrinos of significantly higher energies compared to other proton-proton chains, but they are far more common in stars more massive and hotter than the sun. These two processes do not contribute significantly to the sun's energy output.

- e) The pp-I chain has an energy output of 26.73 MeV but loses 0.59 MeV of that energy for every ${}^4_2\text{He}$ produced, and it accounts for 83.3% of the alpha particles produced. The pp-II chain has an energy output of 19.8 MeV and energy loss of 0.81 MeV, and it accounts for 16.7% of the alpha particles produced. Assuming that the neutrino does not interact significantly with other particles in the interior of a star, what is the energy contribution to the luminosity of the star for every alpha particle produced?
- f) The electron scattering cross section of a neutrino can be thought of as the surface area that an electron occupies. This is given as

$$\sigma \approx 10^{-48} \left(\frac{E_\nu}{m_e c^2} \right)^2 \text{ m}^2.$$

The number density of electrons in the sun is on the order of 10^{-30} m^3 and the rest energy of an electron is on the order of 1 MeV. Estimate to an order of magnitude the mean free path of a neutrino with energy of 1 MeV in a medium with the same density as the sun? Is the assumption we made in the previous part valid?

2 Applied Problems

Solar Sails (30 pts.)

5

Light from the sun exerts radiation pressure on conductors due to the photoelectric effect. One concept for spacecraft design that would eliminate the need for propellant would use the radiation pressure from the sun to increase the energy of the probe, allowing it to visit other planets in the solar system. In this problem we investigate the possibility of using solar sails to propel spacecraft.

- a) Find the force due to radiation pressure on an ideal sail of area A , distance r from a star with power output P , and facing the sun, assuming that the photoelectric effect occurs in this material and that all light is reflected from the sail. Express your answer in terms of A , r , P , and/or any fundamental constants.
- b) A solar sail of mass m and area A orbits a star of mass M in a completely circular orbit of radius r_1 . The sail uses radiation pressure to propel itself to a distance of r_2 . What is the change in kinetic energy of the sail in terms of P , m , M , A , r_1 , r_2 , and/or any fundamental constants?
- c) Now suppose the sail is not perfectly reflecting and that the energy reflected by the sail E_r and radiant energy that reaches the sail E_i always follows $E_r = nE_i$. If the sail reaches a certain temperature T_m , it will burn up. Find the minimum value of r_1 so that the sail is not at risk of burning up (the equilibrium temperature remains lower than T_m). Express your answer in terms of P , n , m , M , A , r_2 , and/or any fundamental constants.
- d) A solar sail of mass m and area A is directly facing the sun. At $t = 0$, the sail is in a circular orbit with no radial velocity. Assume the intensity of light I hitting the sail is constant and that $\Delta r \ll r_0$, where Δr is the change in radius in time t . What is the radial velocity v_r in terms of A , m , I , t , and/or any fundamental constants.
- e) Now find v_r from the previous problem, assuming that the star has power P and that the sail begins at radius r_0 from the star. Assume that $\Delta r \ll r_0$. You may not assume that I is constant. Express your answer in terms of r_0 , P , t , m , A , and/or any fundamental constants.

Although solar sails are a promising technology with the potential of cutting costs for interplanetary exploration by light-weight, unmanned probes, the acceleration of these sails remains much lower than that of spaceships using conventional propulsion methods. For that reason, solar-powered spacecraft would be too slow for most purposes.

Hydrostatic Equilibrium (60 pts.)

6

Gravitational forces inside stars are balanced primarily by pressure gradients. Consider a star of mass M , radius R , and uniform mass density.

- a) Considering the forces on a thin spherical shell at distance r from the star's center and thickness dr , determine how pressure inside a star varies according to distance.
- b) Use your result from part a to determine the pressure at the center of the star.
- c) Suppose a disturbance increases the pressure at the star's center by ΔP . Find the new gravitational field inside the star at a distance r from the star's center.
- d) Consider the initial star, before the disturbance from part c. Assuming the gas in the star is ideal, derive an estimate of how the temperature inside the star varies according to the distance to the star's center.

The hydrostatic equation derived in part a can also be used to analyze the atmosphere of stars. Consider a star whose atmosphere consists of an isothermal ideal gas of temperature T , where the gas molecules have mass m .

- e) Suppose the air pressure at the star's surface is P_0 . Find the air pressure at a height h above the star's surface.

In reality, most star atmospheres are not constant in temperature. Hotter air tends to rise in the atmosphere and push colder air to the star's surface, a process known as convection. Suppose these convective processes are adiabatic, where the gas molecules have adiabatic constant γ . Also, assume that these processes happen very slowly.

- f) Suppose the temperature at the star's surface is T_0 . Find the air temperature at a height h above the star's surface.
- g) Our sun's atmosphere is composed primarily of diatomic hydrogen gas. Given that the surface temperature of the sun is approximately 5500 K, use your result from part f to estimate the air temperature at a height 100,000 kilometers above the sun's surface.

White Dwarves (60 pts.)

7

When stars run out of hydrogen and helium to fuse, they undergo core collapse due to gravity. In lower-mass stars like the sun, electron degeneracy pressure builds up, which stops the collapse. Electron degeneracy pressure occurs due to Pauli's Exclusion Principle, which states that no two electrons can occupy the same states. Eventually, the star sheds its outer layers and only the core remains, which is known as a white dwarf.

Electron degeneracy pressure must be strong enough to keep the white dwarf "star" in stable equilibrium. Letting the system be the electrons in the white dwarf, the total energy consists of the gravitational binding energy and the kinetic energy of the electrons.

- a) The gravitational binding energy is the negative of the energy required to separate each particle to a distance very far away from every other particle. What is the gravitational binding energy of a white dwarf with radius R and mass M , in terms of R , M , and/or any fundamental constants?

At the high densities inside white dwarf stars, electrons become degenerate, which means they occupy their lowest possible states. The wave number of an electron is $k = \frac{2\pi}{\lambda}$, where λ is the de Broglie wavelength. The number of quantum states below a certain wave number k is given by

$$N = \frac{2V}{(2\pi)^3} \cdot \left(\frac{4}{3} \pi k^3 \right),$$

where V is volume.

For parts b and c, express your answer in terms of N , V , m , m_p , dp , and/or any fundamental constants.

- b) In a system of degenerate electrons with volume V and number of electrons N , find the number of electron states with momenta between p and $p + dp$.
- c) The kinetic energy of a particle of mass m is related to the magnitude of its momentum p by the formula $K = p^2/2m$. What is the total energy of a system of degenerate electrons with volume V and number of electrons N ?

Stars can be approximated as being at constant entropy. The principle of minimum energy is a result of the Second Law of Thermodynamics. It states that a closed system (one that can transfer energy but not matter) at constant entropy will experience equilibrium at the state of lowest total energy.

- d) The sun has a mass of 1.99×10^{30} kg and a radius of 6.96×10^8 m. Find the equilibrium radius of a white dwarf star with mass M in terms of R_s , M_s , and M , where R_s and M_s are the radius and mass of the sun.
- e) As the mass of a white dwarf increases, so does the velocity of electrons. If the velocity of electrons is comparable to the speed of light, find the equation of total energy in terms of M , R , m , m_p (mass of proton) assuming that for all electrons $p \gg mc$.
- f) For some mass M_c , there is no equilibrium radius for the white dwarf. Find this critical value M_c .

A more realistic calculation of M_c accounts for the change in density inside stars and yields $M_c = 1.4M_s$. Once the mass of white dwarfs exceed this critical value, core collapse resumes, which may lead to a supernova and the formation of a neutron star (see Problem 9).

Binary Star Systems (50 pts.)

8

Many of the stars in the universe are not single stars, but instead a pair of two stars that orbit each other. According to some source, 85% of stars are part of binary star systems.

Detecting binary stars is made difficult, and in some cases impossible, due to diffraction and aberration effects. While aberration effects can be reduced and nearly eliminated, diffraction effects are impossible to remove. The angular separation required for two objects to be barely resolvable is proportional to the first power of the wavelength of light.

- a) Estimate to one order of magnitude the minimum angular separation between two stars for them to be resolvable by a telescope one meter in diameter.
- b) The minimum distance between two sources of yellow light (575 nm) located 1 kilometer away from a 10 cm wide telescope for them to be barely resolvable by Rayleigh's criterion is 7.02 mm. Kepler-16 is a binary star system 90 light years from earth that primarily emits light with a wavelength of 600 nm. Calculate the minimum distance between two stars for them to be resolvable by a 1 meter wide telescope.

The separation between the two stars in the Kepler-16 system is 3.29×10^{10} , which is less than the minimum distance for the two stars to appear distinguishable from one another. This is one case in which binary star systems cannot be reliably detected by direct observation.

Another method of detecting binary star systems is through observing Doppler shifts in the spectral lines of the stars.

- c) Astronomers observe that wavelengths of a pair of spectral lines from a possible binary system oscillate with amplitude $\Delta\lambda$, where the two wavelengths oscillate 180° out of phase. The average wavelength of both spectral lines is λ_0 . Assume that the stars pass directly in front of each other when observed from the Earth and that both move in circular orbits. Find the velocity of the two stars in terms of λ_0 , $\Delta\lambda$, and/or any fundamental constants.
- d) If the period of the oscillations in the previous part is T , what is the mass of each of the two stars? Express your answer in terms of λ_0 , $\Delta\lambda$, T , and/or any fundamental constants.
- e) Binary star systems can also be detected using the transit method, in which one star passes directly in front of or behind the other star, changing the total brightness. This method is particularly useful when one star is much more luminous (the power output is much greater) than the other.

Suppose a star system of a main sequence star and a white dwarf orbit around each other in circular orbits. The sum of the orbital radii is d . Every interval of time T , a primary transit, in which the white dwarf passes in front of the main sequence star, occurs. This causes the intensity of light from the star system to decrease by ΔL . Let the intensity of light when there isn't a transit be L_0 , and the mass densities of the main sequence star and white dwarf to be ρ_1 and ρ_2 . During the secondary transit, when the white dwarf passes behind the main sequence star, no change in brightness is observed. What is the mass of the larger star in terms of d , L_0 , ΔL , ρ_1 , ρ_2 , T , and any fundamental constants?

Neutron Stars and Supernovae (60 pts.)

9

This problem is divided into three parts. Each part can be solved independently. You may find the following information useful: the mass of the sun is 1.99×10^{30} kg, and the gravitational constant is $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$.

When a white dwarf exceeds the Chandrasekhar limit, collapse due to gravity resumes. This can occur in binary star systems, in which a white dwarf star gains mass from its stellar neighbor, or in massive stars, where the core becomes too massive to maintain stable equilibrium. As the core collapses, temperature increases, which allows for the fusion of increasingly heavy nuclei. This stops the core collapse temporarily until the material for fusion is used up.

Eventually, iron builds up in the stellar core. The fusion of iron is endothermic, so as other elements are used up in fusion processes, nuclear fusion is unable to support the weight of the star. Core collapse resumes again. In some stars, due to processes that are not yet completely understood, the core rebounds, leading to a supernova and the formation of a neutron star. In other stars, the core continues to collapse into a black hole. In this problem, we investigate the first scenario.

- a) During core collapse, as temperature increases, protons can undergo spontaneous electron capture, forming neutrons.

- i Normally, free neutrons decay according to the equation

$$n \longrightarrow p + e^- + \bar{\nu}_e.$$

Neglecting the masses of the electron and antineutrino (which are negligible), and using $m_p = 1.67262 \times 10^{-27}$ kg, and a neutron, $m_n = 1.67493 \times 10^{-27}$ kg, find the energy E_0 released in this reaction.

- ii Using quantum statistics (also seen in Problem 7), the maximum momentum for a group of N degenerate electrons is

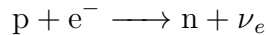
$$p_{\max} = \frac{(3\pi^2)^{1/3} h}{2\pi} \left(\frac{N}{V} \right)^{1/3}.$$

The electrons are highly relativistic, so you may assume that $p \gg mc$, where m is the electron rest mass, p is the electron momentum, and c is the speed of light. The mass of an electron is 9.11×10^{-31} kg. At what electron number density n is the maximum energy of the electrons equal to the energy E_0 ?

- iii Using your answer for part aii), and assuming there is an electron for 2 nucleons (proton or neutron), find to one significant figure the mass density for the condition in the previous part.

During core collapse, energy is used up in the photodisintegration of iron nuclei into alpha particles and alpha particles into protons and neutrons. Once

the mass density exceeds the value found in aiii), neutron decay is no longer spontaneous. Instead, the reverse reaction



becomes possible and dominates in the star. As a result, the majority of protons in the core are converted to neutrons through electron capture.

This collapsed stellar core is now a neutron star. Neutron degeneracy pressure, which works in a similar way as electron degeneracy, is then able to stop the core collapse for stars below the Oppenheimer-Volkoff limit (estimated at between 2 and 3 solar masses).

- b) The time scale of the core collapse depends only on the gravitational constant G and the density of the star. Assume the Chandrasekhar mass is 1.4 solar masses and the radius of a white dwarf is on the order of 1000 km. Estimate to an order of magnitude the time scale of the core collapse.
- c) While a neutron star is being formed, the star blasts its outer layers in a extremely powerful explosion known as a supernova. These explosions are the most energetic events known in the universe. Although the exact mechanisms of supernova explosions are not well known, one can derive the energy budget of a collapsing star.
 - i The energy released in a supernova explosion is the change in gravitational binding energy. The radius of a neutron star is around 20 km and the radius of a white dwarf star near the Chandrasekhar limit is around 10000 km. The Chandrasekhar mass is around 1.4 solar masses. Find to one significant figure this change in gravitational binding energy.
 - ii Around 10% of this energy powers the conversion of protons and electrons to neutrons. Some energy is used to unbind the stellar envelope, which is the stellar material not included in the neutron star, from the core. Assume that the binding energy of the stellar envelope to itself is on the same order of the initial binding energy of the star. What proportion of the total released energy found in part ci) is converted to unbinding the stellar envelope?
 - iii Some energy is converted to kinetic energy of the stellar envelope. By observations, the speed of the ejecta is around 10000 km/s. For a star with total mass of 10 solar masses, find to one significant figure the percentage of the star's total energy (part ci) that is converted into kinetic energy of the stellar envelope.

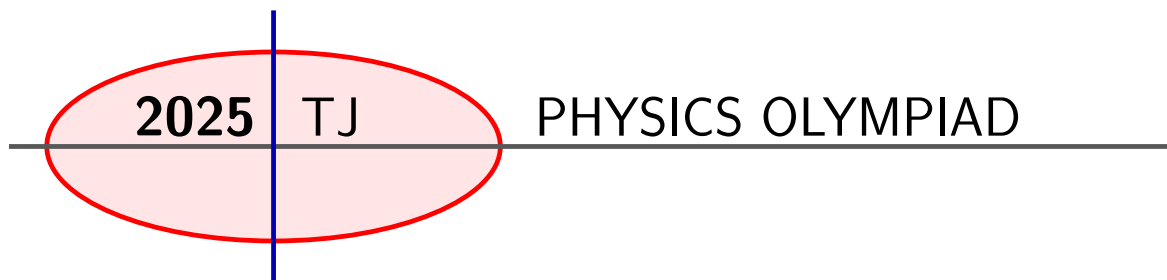
You should have found that the formation of neutrons, unbinding of stellar envelope, and kinetic energy of the stellar envelope requires only a small portion of the energy released by the core collapse. The energy carried away by radiation also use only a small amount of the available energy. It is believed that neutrinos carry away the remaining energy.

While we know that energy must be transferred from the core to the stellar envelope, research is still ongoing to determine how this process occurs (see further resources section).

3 Further Resources

For those interested in further resources, here is an enumeration of articles and web pages regarding current equilibria applications and news, as well as resources for learning about equilibria. Click on the hyperlinks for more information.

- [Pulsars](#)
- [Application of Parallax in Measuring Distances in the Solar System](#)
- [Heat Transfer from Radiation in the Sun](#)
- [Analysis on Radiation Pressure Effects on Spacecraft](#)
- [Cherenkov Effect \(Used in Neutrino Detection\)](#)
- [Composition of the Sun](#)
- [Stellar Life Cycle](#)
- [End of the Sun's Life Cycle](#)
- [Helium Flash](#)
- [Black Holes](#)
- [Supernova](#)



2025 TJ Physics Olympiad: Motion

- The contest lasts from **June 14th, 2025** 12 A.M EST to **June 16th, 2025** 11:59 P.M EST.
- All solutions must either be typeset in L^AT_EX/Typst (preferred), or clearly and legibly handwritten. Illegible handwriting will not receive any points (keep in mind cursive can be hard to read as well).
- Solutions should mention the concepts used (e.g. "by conservation of energy"), but please don't write a 5-paragraph essay.
- The usage of Generative AI is PROHIBITED (and will not be useful). Usage can result in disqualification.
- You MAY use online software including, but not limited to, Wolfram Alpha, Matlab, Python, or Julia, solely as a mathematical aid.
- All questions are equally weighted, but not necessarily the same level of difficulty. The subparts may not be equally weighted. Also, the problems may not be given in order of difficulty, so make sure to try all of them.
- Even if you are not able to fully solve a problem, please write up what you can. We want to see how you approached the problem, and partial credit may be awarded.
- Join [our discord](#) to ask questions
- If you need to contact us for clarifications, please email us at tjhsstphysics@gmail.com
- Yes, this cover was designed to look like the cover of the $F_{\text{net}} = ma$ exam.

We acknowledge the following people for their contributions to this year's exam (in alphabetical order):

*Sanchali Banerjee, Tiger Deng, Aarush Deshpande, Jason Hao, Sophia Hou, Kabeer Parmar, Ryan Singh,
and Eric Xie*

Introduction

From the earliest water wheels that harnessed river currents to the jet and rocket engines that propel us skyward, motion and energy has powered our machines and understanding of nature. In the atmosphere, heat differentials drive ocean currents, and in engines, chemical and thermal energy convert into mechanical work. Conservation laws, fluid dynamics, thermodynamic efficiency, and statistical behavior, govern how energy is stored, transferred, and transformed into motion.

TJ Physics Team is proud to announce the seventh annual TJ Physics Olympiad (TJPhO)! We hope that our problems deepen your appreciation for how motion and energy intertwine in every "engine", whether natural or engineered, and inspire you to apply these principles in your own problem-solving.

Happy exploring!

TJ Physics Team Officers

A Relativistic Rocket

A rocket is launching from Earth, with a constant velocity v , and ejects photons to produce thrust (cue the *Little Einsteins* theme song). Assume it is a hot day, and the index of refraction for light is given by $\eta = n_0(1 + ay)$, where $a = 1.5 \times 10^{-6} \text{ m}^{-1}$, and y is the distance from the surface of the earth. Ignore the curvature of the Earth. Unless stated otherwise, all quantities are measured in the ground frame.

- A) Assuming the rocket launches at an angle θ from the normal, find an approximate expression for the distance $d(t)$ (where $t = 0$ corresponds to rocket launch) away from the rocket you must stand to be able to see the rocket when its height is $h \sim 1 \text{ km}$ (see Fig 1). Neglect the height of the person, and the light emitted by the Sun. You may also assume the observer can only see a tiny bit above the horizontal, and that the height of the rocket is such that the light becomes close to parallel at the ground level.

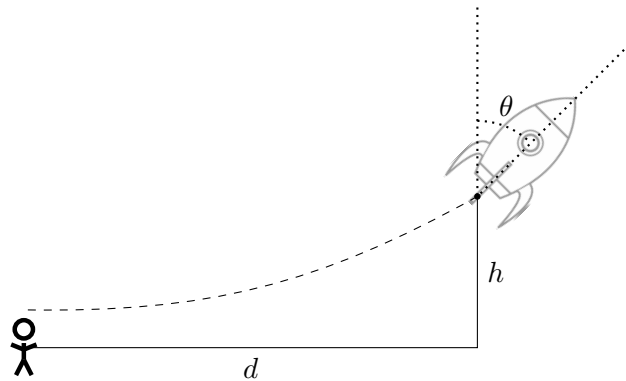


Figure 1

- B) Repeat the previous problem, assuming the observer can look at any angle $\leq 90^\circ$ from the horizontal. If your answer involves hyperbolic trig functions, you should correctly approximate your solution down to trigonometric functions for full credit.
- C) Assume the observer maintains the distance d , and the rocket is at the same order of magnitude of h as in the previous part. If the rocket has a proper length L , what is the length of the rocket in the observers frame?

Solar Panels

Solar Panels are made out of photovoltaic cells, which are semiconductors. When photons hit the cells, they get absorbed. Then, they can excite electrons in the cell effectively only if their energies are greater than the band gap energy. If this is the case, the excited electrons jump from the valence band to the conduction band, where they can freely move through the semiconductor.

- (A) When photons excite electrons in a photovoltaic cell, they generate a photocurrent, producing electricity. In real photovoltaic cells, about 90% of the excited electrons can move freely in the conduction band.
- i) Find the number of photons that hit the solar panel per unit area every second with energies high enough to excite electrons. Assume the sun is a blackbody with temperature $T = 5800\text{K}$ and that the solar panel is a closed sphere surrounding the sun on the sun's surface. You can also assume the minimum wavelength of light that reaches the solar panel is $\lambda_{\min} = 280\text{nm}$ and a band gap energy of 1.1 eV .
 - ii) Find the magnitude of the current that is generated in a realistic (non-idealized) semiconductor.
- (B) Photovoltaic cells themselves consist of two layers of semiconductor materials, called p-type and n-type semiconductors. Silicon, which is typically used in semiconductors, can be "doped" by replacing a small amount of the atoms, usually about 1 in 10^7 , with those of another element. If silicon is "doped" with phosphorus, it creates an n-type semiconductor and an excess of electrons. However, if silicon is "doped" with aluminum, a p-type semiconductor is created, and an excess of "holes" (which represent the absence of electrons) is maintained. These p-n junctions in photovoltaic cells create diodes. So, we will model the solar cells as ideal diodes.
- i) Assuming the cell is at a temperature of 300K , find the maximum theoretical voltage across the diode. Model the energy transfer between the sun and the solar panel as a thermal energy transfer.
 - ii) Find the maximum power this photovoltaic cell can generate in this situation, in terms of the power P emitted by the sun.
- (In reality, most solar panels have an efficiency of around 30% of their maximum power production possible).

Light Bouncing

We have two fixed, perfectly reflective walls, with the distance between them being 2ℓ . A double-sided mirror with mass m (which is lined up with the walls) is placed in the center between the two walls. On each side of the mirror, light with energy E_0 is placed, which bounces back and forth between the mirror and the walls. The mirror is given a speed v_0 to the left. Assume $v_0 \ll c$, that v_0 is insufficient to reach the left wall, ℓ is much greater than the wavelength of the light, and that there is no energy dissipation. The mirror will oscillate between the two walls.

- (A) Find the energy of the light to the left of the mirror when the mirror is closest to the left wall.
- (B) Estimate the period of the oscillations. Assume that the maximum displacement $A \ll \ell$.

Fluids in Action

In this problem, we aim to demonstrate how one arrives at some common equations used for modeling fluids, used when designing hydropower. Note that this question will require knowledge of Vector Calculus. If you need to learn it, <https://www.damtp.cam.ac.uk/user/tong/vc.html> is excellent, and so is the first chapter of Griffiths Electrodynamics.

- A) Just to get in the fluids' mindset, let's start with some problems that should be approachable without significant fluids' knowledge.
- i) Suppose we have a can of base area A and height H , and is filled with water of density ρ . A small hole of area $A_0 \ll A$ is then opened at the bottom of the can. At the same time, a piston pushes down on the top of the water with a constant force F . Find the height of the water in the can as a function of time.
 - ii) Repeat the previous problem, without the $A_0 \ll A$ assumption.
- B) Now, we shall derive some important equations in fluid dynamics! Note that we shall often work with a velocity field $\mathbf{v}(\mathbf{x}, t)$, and we define the operator D/Dt (called the material derivative) as

$$\frac{D}{Dt} := \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla$$

In essence, it is a way of relating the change in a Lagrangian quantity, which is a property observed by a moving fluid particle, to an Eulerian quantity at a fixed point in space.

- i) The rate at which mass is moving through a surface is given by

$$\frac{DM}{Dt} = \frac{d}{dt} \int_V \rho dV + \oint_{\partial V} \rho \mathbf{v} \cdot d\mathbf{S}$$

where ∂V denotes the boundary of V , and ρ is the density of the fluid. Show that

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{v}$$

This is called the continuity equation, and it was first obtained by Euler in 1753. Note that while we derived it in the context of fluid dynamics, it can also be applied to other quantities, such as electric charge.

- ii) By applying Newton's Second Law $\mathbf{F} = D\mathbf{P}/Dt$ to a volume, show that

$$\rho \frac{D\mathbf{v}}{Dt} = \rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = f^*$$

where f^* is the effective force density (force per unit volume). This is called Cauchy's equation, and in conjunction with the continuity equation it governs the behavior of *all continuous matter*! Hint: what are all the contributions to the change in momentum?

- iii) For fluids without any viscosity, the only forces at play are gravity and pressure. Under this assumption, find f^* and substitute it into Cauchy's equation. This is called *Euler's Equation for ideal fluids*.
- iv) A fluid is in steady flow when the flow doesn't change with time. Show that for an incompressible fluid with steady flow and constant density ρ_0 , the quantity

$$H := \frac{1}{2} v^2 + \phi + \frac{p}{\rho_0}$$

where ϕ is a kind of effective potential, is conserved along streamlines. This is called Bernoulli's theorem. Show that this is a generalization of the typical Bernoulli's equation

$$P_0 + \frac{1}{2}\rho v^2 + \rho gh = \text{const}$$

- C) Having introduced some common ideas from fluid mechanics, we now turn to a familiar topic: waves! Waves controlled by pressure and gravity are called *gravity waves*, and those driven by pressure and surface tension are called *capillary waves*. In an interface between two fluids of similar density, such as a saline layer of the sea overlaid with a brackish layer, *internal gravity waves* may also arise.

Unfortunately, the more unusual types of waves originating from the non-linear aspects (the advective term) of fluid mechanics, such as the run-up of waves on a beach or a sonic boom, require a much more challenging amount of mathematics. Instead, we shall focus on applications involving small-amplitude surface waves.

- i) As a fun thought, consider what would happen if you lifted a mound of water with height a and width λ , and it let splash back down in a characteristic time τ into a sea of depth $d \ll \lambda$. Since the sea is shallow, the ground will force all of the water to move away horizontally with a speed v . Estimate v up to a dimensionless coefficient.
- ii) Small amplitude, inviscid gravity waves in incompressible water obeys

$$\frac{\partial \mathbf{v}}{\partial t} = \mathbf{g} - \frac{1}{\rho_0} \nabla p, \quad \nabla \cdot \mathbf{v} = 0 \quad (1)$$

In other words, the time derivative of the velocity is a gradient field, implying that the velocity takes the form of a gradient field plus a time independent field (such as one containing vorticity). However, for waves, we don't really care about the time independent field so we shall claim

$$\mathbf{v} = \nabla \psi$$

which, due to the incompressibility condition, must satisfy Laplace's equation $\nabla^2 \psi = 0$. Find the general form solution for p in terms of ψ and other given variables in Equation 1.

- iii) Assuming the water is surrounded by air (or a vacuum) above the surface, and that the amplitude of waves a very small ($a \ll d, \lambda$ to be precise), come up with two boundary conditions for the general form solution derived above.

Hint: it may be helpful to consider the water as a surface $z = h$ for some h .

Further analysis of shallow and deep water waves consists mostly of assuming that ψ takes the form of an elementary harmonic wave, and applying the above boundary conditions to derive a result. As such, we decided not to include it in the olympiad, but it's very interesting and we highly recommend reading about it!

For example, if you're curious about capillary waves, the effect of surface tension on deep water waves is to increase the gravitational acceleration to

$$g' = g_0 + \frac{\gamma k^2}{\rho_0}$$

where γ is the surface tension, and k would be determined by a boundary condition.

Hurricanes

- A) Suppose that in a hurricane, the ocean has a constant temperature 30°C and the ocean is in thermal equilibrium with the air directly above it. The air temperature in the upper troposphere is maintained at -55°C from conduction with the upper atmosphere. If in a small time dt , the ocean rate of heat transfer from the ocean to the atmosphere is $\frac{dQ}{dt}$ and the rate of change of the kinetic energy of the air is $\frac{dK}{dt} = \alpha \frac{dQ}{dt}$, calculate numerically the maximum possible value of α .
- B) In a rotating reference frame with angular velocity $\vec{\Omega}$, the coriolis force on a particle with mass m , velocity $\vec{v}$, and displacement $\vec{r}$ from the axis of rotation is $\vec{F}_{cor} = -2m\vec{\Omega} \times \vec{v}$, and this causes hurricanes to rotate counter-clockwise in the northern hemisphere and clockwise in the southern hemisphere. For a small volume of air moving directly east in a hurricane at 30° latitude in the northern hemisphere, let v_0 be the required velocity of the air particles with respect to the Earth for the component of Coriolis acceleration parallel to the ground to be comparable to the gravitational acceleration. You may assume that the air is close to the Earth's surface. Find numerically v_0 to one significant figure. The radius of the earth is 6.37×10^6 m.
- C) Now we will consider a more formal derivation of the coriolis force in the general case for a hurricane. Again, consider a hurricane at a latitude of 30° in the northern hemisphere, but now with a wind speed much less than that found in the previous problem ($v \ll v_0$). Consider a model of the hurricane as a flat circle with radius much smaller than the Earth's radius, and in polar coordinates for any location on the circumference of this hurricane, let θ be the angle counterclockwise from North. Find the function $F_{\parallel}(\theta)$ in terms of Ω as defined in the previous problem, v , and/or any fundamental constants (where $F_{\parallel}$ is the component of the Coriolis force in the radial out direction and parallel to the ground). You may assume that the air in the hurricane always maintains the same height above the ground.
- D) Suppose a hurricane has an eye of radius r_1 and an outer radius of r_2 . The air in the hurricane at sea level has a constant density ρ , and the angular velocity of the Earth's spin is Ω . If the difference between pressure at distance r_1 from the hurricane's center and pressure at a distance r_2 is $-\Delta p$, where $\Delta p > 0$, find the wind speed at sea level at the outer edge of the hurricane v_0 in terms of ρ , r_1 , r_2 , Ω , Δp , and/or any fundamental constants. You may neglect the viscosity and any drag forces in the air, and assume that a single parcel of air always has a small radial velocity v_r such that $v_r \ll v_0$.
- E) For a hurricane with inner radius $r_1 = 5$ km, outer radius $r_2 = 800$ km, and height $H = 10$ km, consider the function $v(r)$ for $5\text{ km} < r < 800\text{ km}$ (the wind speed as a function of distance from the hurricane's center). The absolute value of the pressure difference between that of $r = r_1$ and $r = r_2$ is $\Delta p = 3000$ Pa. This function can be written as $v(r) = Ar^\alpha$.
- i) If $v(r)$ and r are both in SI units, find numerically α and A , and express A to three significant digits with correct units.
 - ii) Find the maximum wind speed for $5\text{ km} < r < 800\text{ km}$. The density of air at sea level is given as $\rho = 1.225\text{ kg/m}^3$

Jittery Atoms

A small cylinder in a model heat engine contains N identical, non-interacting atoms that can occupy a discrete set of energy levels. As the piston moves, the gas exchanges heat with a high-temperature reservoir at T_H during the "intake" stroke and with a cold reservoir at T_C during the "exhaust" stroke. Under the assumption that between strokes, the gas reaches thermal equilibrium, the macroscopic state of the gas at any instant is specified by the set of occupation numbers,

$$n_1, n_2, \dots, n_m, \quad \sum_{j=1}^m n_j = N, \quad \sum_{j=1}^m \varepsilon_j n_j = E,$$

where ε_j is the j -th single-atom energy level. Boltzmann argued that the entropy of the gas is

$$S = k_B \ln \Omega$$

where Ω is the number of distinct microstates corresponding to the given occupation numbers:

$$\Omega = \frac{N!}{n_1! n_2! \cdots n_m!}$$

Since S is maximized (for fixed total energy E and total atom number N), the most likely occupation numbers $\{n_j\}$ satisfy

$$n_j = A e^{-\beta \varepsilon_j} \quad (\text{canonical distribution}),$$

where $\beta = 1/(k_B T)$ and A is a normalization constant (related to the chemical potential).

In this problem, you will derive the Boltzmann distribution and explain why n_j takes the form $A e^{-\beta \varepsilon_j}$, and then apply these results to a concrete system.

A) Derivation of the Boltzmann Distribution

- i) Determine how to maximize $\ln \Omega$ under $\sum_j n_j = N$ and $\sum_j \varepsilon_j n_j = E$ by using Lagrange Multipliers and/or taking a derivative with respect to n_j and showing that

$$\frac{\partial}{\partial n_j} [-\ln(n_j!) + \alpha n_j + \lambda \varepsilon_j n_j] = 0.$$

- ii) Determine why $n_j = A e^{-\lambda \varepsilon_j}$, and find a simplified approximation for entropy.

B) Application: Four-Level Gas Molecule

The working gas has four energy levels, $\varepsilon_1 = 0$, $\varepsilon_2 = \varepsilon$, $\varepsilon_3 = 3\varepsilon$, $\varepsilon_4 = 5\varepsilon$.

- i) Determine the entropy associated with this system if there are 0.8×10^{10} atoms in the first state, 0.5×10^{10} in the second state, 0.4×10^{10} and 0.3×10^{10} in the fourth.
- ii) What is the maximum entropy possible for this system of four states if the total energy is not fixed?
- iii) Determine the occupation numbers of each of the states if the total energy is fixed at $3 \times 10^{10} \varepsilon$. Give your answers in terms of $x = e^{-\beta \varepsilon}$.
- iv) Determine the temperature of the system under the assumptions of part 3). Give your answer as a decimal multiple of $\frac{\varepsilon}{k_B}$.
- v) What is the maximum entropy possible for this system under the assumptions in part 3)?

A Powerful Piston

A vertical piston of mass m and cross-sectional area A slides without friction in a cylindrical chamber filled with n moles of an ideal diatomic gas. The piston moves without friction and is free to oscillate vertically. The entire system is thermally insulated, so any motion of the piston results in adiabatic compression or expansion of the gas. At equilibrium, the piston is at a height h_0 above the base of the cylinder and the pressure inside the chamber exactly balances the weight of the piston and atmospheric pressure P_0 .

- (A) Determine the equilibrium pressure P of the gas and use it to express the equilibrium temperature T_0 of the gas in terms of the given variables.
- (B) Now suppose the piston is displaced upward by a small amount x and then released. Find the angular frequency of the oscillation of the piston.
- (C) Suppose the piston is now released from rest at a height $h = h_0 + \Delta h$, where $\Delta h \ll h_0$. Assuming the system behaves as a harmonic oscillator, derive an expression for the total mechanical energy of the system in terms of the variables above, and identify the physical origin of this energy in the thermodynamic context.

Some Random Problems

We just had some other ideas that we didn't want to go to waste. They don't fully fit in with the theme, but we think they're fun!

- A) n identical bricks of uniform density and length L are stacked, one above the other, near the edge of a table.
- What is the maximum length the farthest edge of the top brick can protrude over the edge of the table? (Hint: it is related to the n th harmonic number).
 - The table then starts to rotate with an angular velocity ω . Let coefficient of friction between the bricks be μ , and assume the table is made of bricks, and has a radius R . Find the minimum μ such that the maximum distance is the same as that in part (a).
- B) Suppose there is a hemispherical sprinkler with negligible (as in not affecting the range of the water), but nonzero radius r , centered at the origin. It sprays water from its surface uniformly, i.e. the height (volume of water per surface area on the sprinkler is height) of water shot from any point on its surface is the same at H_0 . The water shot from a point is shot along the line from the center of the hemisphere to said point, at a speed v .

Find a formula in terms of x (with x ranging from 0 to the maximum range) for the distribution of the height of the water deposited at any point along the positive x -axis. You should approximate to the first degree when applicable. (Hint: Integrating your formula from $x = 0$ to $x = \frac{v^2}{g}$ should give you $\frac{\pi H_0 r}{2}$)

Wording clarification: Note that the sprinkler deposits a volume of water over an area of ground, so each point on the ground receives a "height" of water. The height here has nothing to do with the trajectory of the water, and instead is a measure of how "wet" the ground is, like how weather stations report "We have had 1 inch of rain today".

Another Random Problem

Here's another problem which doesn't fully fit into the theme, but we think is quite interesting.

In this problem, we shall derive the shape of an axially symmetric pending droplet (think of a cartoon raindrop). Note that due to the axial symmetry, we shall work with cylindrical coordinates, with the origin at the bottom of the droplet.

If you're curious, we can create a droplet of this kind by taking for example, a pipette with a rubber bulb allowing us to vary the pressure. When the bulb is squeezed, a droplet emerges and will fall once it becomes large enough.

- i) First, find the critical radius R_c , which is defined as the radius for which the hydrostatic pressure across a spherical drop with surface tension γ is equal to the pressure excess due to surface tension.

As an aside, for the effect of gravity to be negligible in a spherical droplet, its radius must be $a \ll R_c$.

- ii) Show that given the arc length s of the droplet surface as viewed in the rz plane, and the angle of elevation θ of ds , we must have

$$\frac{dr}{ds} = \cos \theta, \quad \frac{dz}{ds} = \sin \theta$$

- iii) Find the two radii of curvature of the raindrop R_1 and R_2 , where R_1 the radius of the circle in the rz plane.

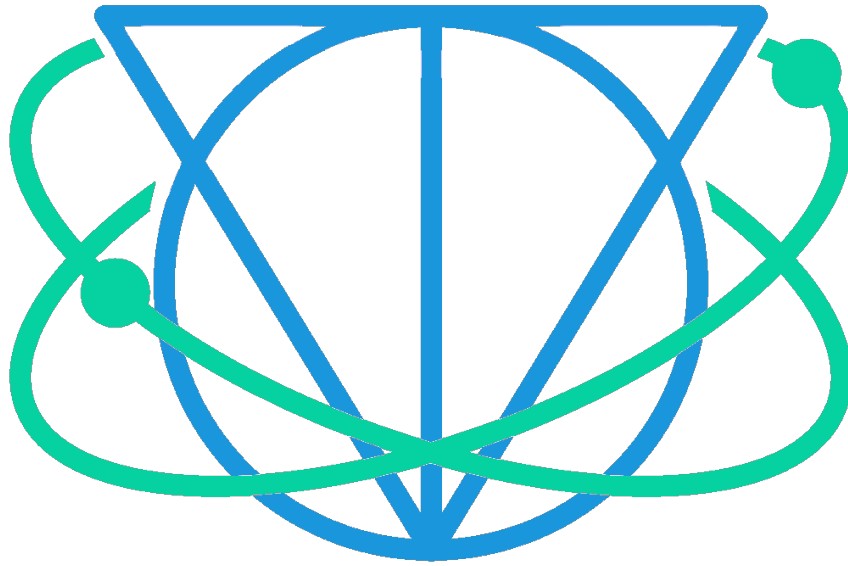
Hint: the second center of curvature lies on the z axis

- iv) Young-Laplace's equation states that

$$\Delta p = -\gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

Note that a contribution to the pressure discontinuity is always positive on the side of the surface containing the center of curvature, otherwise negative. Find an expression for $d\theta/ds$ in terms of θ , r , z , R_c , and R_0 , where $R_0 = R_1 = R_2$ at $r = z = \theta = 0$

We now have a set of differential equations that can be solved numerically for the shape of the droplet, given a value of R_0 ! If you'd like to explore some more, you can run some simulations with different R_0 values. If $R_0 > R_{01} = 0.778\dots$, the radial distance grows monotonically with s , but if it's less than R_{01} it's shaped like an old-fashioned bottle with multiple waists. However, note that these simulations won't earn you any points on TJPhO :P



2026 Thomas Jefferson Physics Olympiad

We acknowledge the following people for their contributions to this year's exam (in alphabetical order):

Aarush Deshpande, Eric Xie, Kabeer Parmar, Noah Kops, Tiger Deng

Instructions

This year's contest lasts from **June 27, 2026 12A.M. EST** to **June 30, 2026 12 A.M. EST**.

- Please join the discord for the latest exam updates: <https://discord.gg/gATxvtx9R>.
- All clarifications should be emailed to tjhsstphysicssteam@gmail.com
- Use $g = 9.8 \frac{\text{m}}{\text{s}^2}$ **unless otherwise specified**.
- Questions are not necessarily in order of difficulty, so try all of them!
- The final solution should be a write-up, clearly detailing all the steps to reach a final solution, followed by a **boxed** final answer.
- Handwriting counts - if it is illegible, it is not being graded. If you are unsure if your handwriting is legible, typeset your work with LaTeX or Typst. Keep in mind that cursive is also hard to read.
- Answers or steps that are not dimensionally correct shall receive 0 points.
- Working that is difficult to follow shall receive 0 points.
- You are welcome to use Computer Algebra Systems, including (but not limited to) SymPy, Wolfram, and Matlab.
- Generative AI is **strictly forbidden**, and any usage will result in disqualification. This competition is just for fun, please don't ruin it.
- **Do not communicate information to anyone except your team members before the end of the competition**

Notation Review

- The symbol c will be used to denote the speed of light, unless otherwise stated.
- An average of a quantity v is denoted as follows: $\langle v \rangle$
- The notation ∂F denotes the boundary of a manifold F . For example, if S is a surface then ∂S is the curve that is the boundary of the surface. Likewise, if V is a volume, then ∂V is the surface of the volume.
- For flux integrals, we may use $d\mathbf{S} \equiv \mathbf{n} dA$, where $\mathbf{n}$ is the assumed well defined normal vector at each point.
- For volume and surface integrals, we will simply use one integral, i.e. $\int_V F dV$ instead of $\iiint_V F dV$.

Mathematical Prerequisites

Problems marked with a [A] may require some of the following theorems from multivariable calculus:

$$\begin{aligned}\int_a^b \nabla f \cdot d\mathbf{s} &= f(b) - f(a) \\ \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} &= \int_{\partial S} \mathbf{F} \cdot d\vec{\ell} \\ \int_V \nabla \cdot \mathbf{F} dV &= \int_{\partial V} \mathbf{F} \cdot d\mathbf{S}\end{aligned}$$

where ∇ is the “vector” of partial derivatives in Cartesian coordinates $\langle \partial_x, \partial_y, \partial_z \rangle$, $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a scalar field, and $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a vector field. In general, these problems will be worth less points, and are only included to demonstrate some interesting phenomena.

Problems

1. Marangoni Stresses

In the 1860s, Carlo Marangoni showed that surface tension gradients can actually produce shear forces that we call Marangoni stresses. Such surface tension gradients arise from either gradients in temperature, or changes in the chemical composition at the interface.

- A) [A] Show that a fluid configuration cannot be static if such surface tension gradients exist
- B) Consider a spherical soap bubble of radius R , mass density ρ , and thickness $\tau \ll R$ in a vacuum. An external agent then slowly places the bubble in a uniform gravitational field $-g_0 \hat{z}$. Assuming the bubble's radius stays approximately constant, what is the change in the surface energy of the bubble?

2. Spinny Spinny Dipole

Consider a classical model of the electron as a rotating magnetic dipole with magnetic dipole moment $p_m = e\hbar/2m$. If the dipole's angular velocity is ω , find $\hbar\omega$ in terms of the rest energy of the electron, $E = mc^2$. You should treat the problem classically, i.e. $\gamma \approx 1$ everywhere in space, where γ is the Lorentz factor.

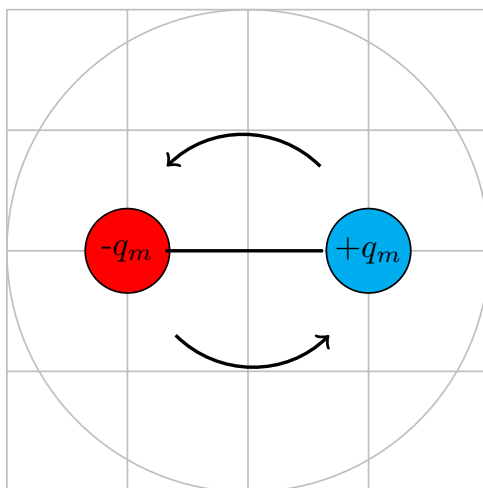


Figure 1: The electron, modeled as a magnetic dipole rotating in the plane of the figure

3. A Dangerous Atmosphere

The atmosphere of a planet consists of Uranium Hexafluoride (UF_6) gas. What is the average height of the gas in the atmosphere, relative to the surface of the planet? You may assume that $g = 9.81 \frac{\text{m}}{\text{s}^2}$ and the temperature $T = 80^\circ \text{C}$ are both approximately constant across the atmosphere, and that UF_6 gas behaves ideally.

4. Spin the Bottle

A cylinder of radius r is filled to a height h with a liquid with density ρ . The coefficient of friction between the liquid and the cylinder is μ . The cylinder is pivoted at the bottom end (Figure 2), and begins to rotate with a constant angular velocity ω .

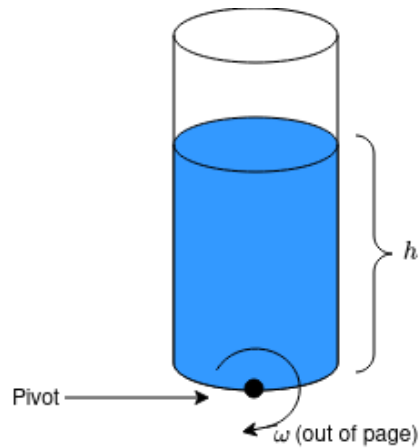
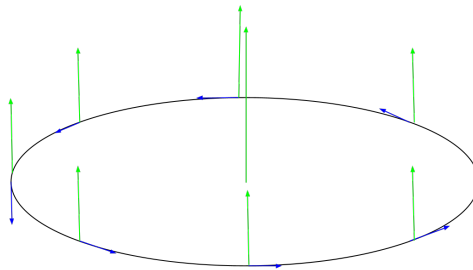


Figure 2: The cylinder is pivoted, and rotates with ω pointing perpendicular to the page

Let the surface tensions of the liquid/air, liquid/cylinder, and cylinder/air interfaces be γ_{LA} , γ_{LC} , and γ_{CA} , respectively. What is the maximum $\omega =: \omega_c$ such that the fluid stays inside the cylinder?

5. Particle Accelerator



Eric wants to construct a simple, circular particle accelerator of radius r to accelerate a particle of charge q and mass m . He plans on generating a time-varying magnetic field within the particle accelerator of average magnitude $\langle B_{\text{within}} \rangle(t) = kt$, using the resulting electric field to speed up the particle and the resulting magnetic field to turn it in an orbit. Assume that the magnetic field is always parallel to the axis of symmetry of the particle accelerator and that the electric field is always pointing parallel to the velocity of the particle. In order for the magnetic field to simultaneously curve the particle and accelerate it correctly, the following condition is required as a consequence of Faraday's Law of Induction: $\Delta B_{\text{loop}} = \frac{1}{2} \Delta \langle B_{\text{within}} \rangle$. Calculate the time derivative of the speed v of the particle as a function of v . Assume $v \sim c$.

6. Extreme Weather

In late August 2005, Hurricane Katrina struck Louisiana. Reports show that near its landfall, the hurricane had maximum wind speeds of around 120 mph (54 m/s) and a radius of maximum wind speed of about 30 miles (48 km). Here, we will model Katrina's inner vortex core using a Rankine vortex model.

Given that maximum tangential wind speed $v_{\theta, \text{max}} = 120$ mph and radius of maximum wind speed $R_c = 30$ miles, and assuming that the maximum tangential wind speed occurs at the outer edge of the vortex core ($r = R_c$)

Derive:

- A) An expression and value for the vertical vorticity inside the vortex core
- B) An expression and value for the time required for an air parcel inside the vortex core to complete one full revolution

7. Suboptimal Induction Heating

Suppose Gyrados constructs a hollow, thin-walled cylinder of radius R and height H without top and bottom faces using a single rectangular sheet of conductive material of thickness $\varepsilon \ll 1$, resistivity ρ , mass m , and heat capacity c_p . If Gyrados places this cylinder in a uniform, time-varying magnetic field of magnitude $B = A \sin(\omega t)$ which is parallel to the axis of symmetry of the cylinder, it will induce a current in the cylinder which will produce heat according to $P = IV$. However, in real life, heating this cylinder would cause it to expand and cause the resistivity to change. The following equations govern how the change in temperature affects the properties of the cylinder: $H = H_0 + \alpha \Delta T$, $R = R_0 + \alpha \frac{R_0}{H_0} \Delta T$, $\rho = \rho_0 + \alpha_1 \Delta T$, and $\varepsilon = \varepsilon_0 + \alpha \frac{\varepsilon_0}{H_0} \Delta T$.

- A) Given initial temperature T_0 , create a differential equation isolating $\frac{dT}{dt}$ on one side, defining it in terms of R_0 , H_0 , ρ_0 , ε_0 , α , α_1 , m , c_p , T , T_0 , A , ω , and t .
- B) Sketch a graph of the solution to this differential equation with labels.

Consider changes in the magnetic field from the induced current negligible.

8. Statistical Physics and Neural Networks

- A) A one-dimensional Ising paramagnet at temperature T consists of N spins in a line. The spins only interact with an external magnetic field and not with each other. The system state is described by a column vector $\vec{\sigma} = [\sigma_1 \cdots \sigma_N]^T$ where σ_i are spin variables (they are ± 1). Its Hamiltonian¹ is

$$\mathcal{H}(\vec{\sigma}) = -\mu_B B \sum_{i=1}^N \sigma_i$$

where B is an external magnetic field and μ_B is the Bohr magneton.

- i) Find the equation of state for $m := \langle \sigma \rangle$, the magnetization per spin.
- ii) Now, we allow the spins to interact with each other by replacing $\mu_B B$ by an effective field, $\mu_B B_e = \mu_B B + Jm$ where $J > 0$ is a coupling between the spins and m is the magnetization. By adding a coupling between spins, we introduce a ferromagnetic state. Find the new equation of state for the magnetization per spin.
- iii) Find the critical temperature T_c for which this system transitions from a paramagnet to a ferromagnet.
- B) There is no requirement for the coupling J to be homogeneous between every pair of spins. In fact, this is not true for many materials. Therefore, we introduce the spin glass Hamiltonian,

$$\mathcal{H}(\vec{\sigma}) = -\frac{1}{2} \frac{\mu_B}{N} \sum_{i=1}^N \sum_{j=1}^N J_{ij} \sigma_i \sigma_j - \mu_B B \sum_{i=1}^N \sigma_i$$

where $J_{ij} = J_{ji}$ and is distributed normally about 0

- i) Find the local effective magnetic field experienced by a particular spin.

¹The total energy

- ii) Find the change in energy ΔE after flipping a given spin σ_i
- iii) Show that the energy decreases monotonically as dynamics continue. Explain why the final configuration is a “frozen” state.
- iv) To characterize the onset of the “freezing,” we introduce an order parameter that measures whether spins remain frozen in time.

$$q = \frac{1}{N} \sum_{i=1}^N \langle \sigma_i \rangle^2$$

q also satisfies the equation,

$$q = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{z^2}{2}\right) \tanh^2(\beta J \sqrt{q} z) dz$$

where $\beta := \frac{1}{k_B T}$. Find the critical temperature T_c below which a non-zero solution for q appears.

- C) We can also ask whether memory can emerge from the collective interactions of simple units. From the parts above, you have shown how the observable macroscopic state of a system is related to the coupling between spins. You have also shown that when these couplings vary between pairs, the system can encode “frozen” states. If we can encode frozen states in the structure of the Hamiltonian $\mathcal{H}$ itself, the system will naturally flow towards these frozen states, or memories. A memory $\vec{\xi}^{(\mu)}$ is indexed by μ out of a total of P stored memories. Now, we construct a network of N “neurons,” and map it to our spin network. If a particular neuron fires, it has $\sigma_i = 1$. Similarly, if it is silent, it has spin $\sigma_i = -1$. In this network, a memory $\vec{\xi}^{(\mu)}$ is just a subset of neurons firing together.
- i) Create a function of the overlap between a network microstate n with the memory μ , $M(\vec{\xi}^{(\mu)}, \vec{\sigma}^{(n)})$
 - ii) When $M(\vec{\xi}^{(\mu)}, \vec{\sigma}^{(n)}) = \pm 1$, the state of the network exactly matches that of the memory (or antimemory), $\vec{\sigma}^{(n)} = \pm \vec{\xi}^{(\mu)}$. Use your overlap function $M(\vec{\xi}^{(\mu)}, \vec{\sigma}^{(n)})$, and the internal energy of the ferromagnet $U = \langle \mathcal{H} \rangle = -\frac{1}{2} J N m^2 - \mu_B B N m$ to construct a Hamiltonian $\mathcal{H}$ for this system in terms of σ and an interaction weight,

$$W_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^{(\mu)} \xi_j^{(\mu)}$$

- D) Now consider the dynamics of such a system. The network evolves in discrete time steps,

$$\vec{\sigma}(t) \rightarrow \vec{\sigma}(t+1)$$

The Hamiltonian also becomes time dependent,

$$\mathcal{H}(t) = -\frac{1}{2} \sum_{i=1}^N h_i(t) \sigma_i(t)$$

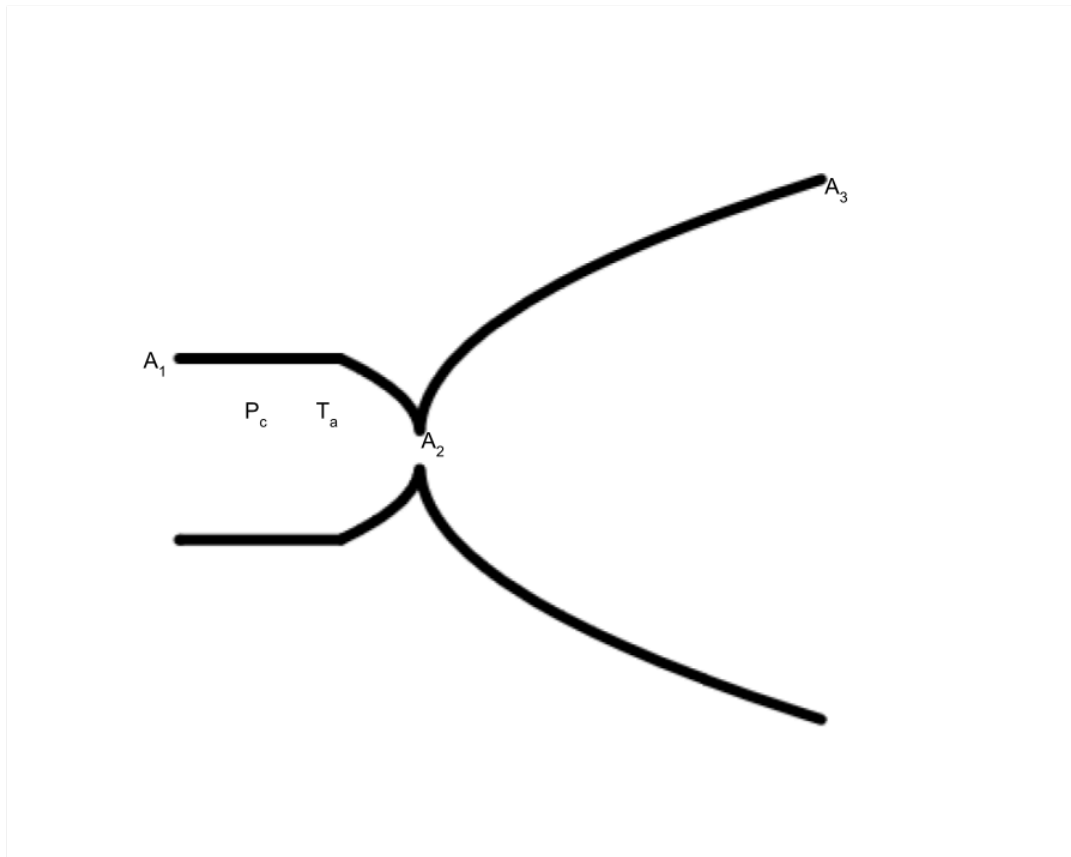
where $h_i(t)$ is the effective field acting on a neuron i ,

$$h_i(t) = \sum_{j=1}^N W_{ij} \sigma_j(t) + \mu_B B_i$$

Now, consider updating a single neuron k . Find the change in the Hamiltonian $\Delta \mathcal{H}(t)$ after one time step in terms of $h_k(t)$ and $\sigma_k(t)$.

- E) A network like this cannot store an infinite number of memories without making mistakes.
- Use your formula for $\sigma_i(t+1)$ from part D to determine the condition under which a memory can be properly recalled by the network.
 - Assuming the patterns are random, and that $N > P \gg 1$, find an approximate formula for the probability that a given memory in P will be recalled incorrectly. Give your answer in terms of N and P .

9. Ethanol Rocket



Bob is thirsty but only has access to 1000kg of 100% pure ethanol ($\text{C}_2\text{H}_5\text{OH}$) (46.07 g/mol), 1389.19036249kg pure oxygen (32.00 g/mol), and a functioning rocket of dry mass m_0 . He decides to combust the ethanol with the oxygen to produce water according to the following balanced equation: $\text{C}_2\text{H}_5\text{OH} + 3 \text{O}_2 \longrightarrow 3 \text{H}_2\text{O} + 2 \text{CO}_2$ which releases $1366.8 \frac{\text{kJ}}{\text{mol C}_2\text{H}_5\text{OH}}$. One mole is equal to $6.022 \cdot 10^{23} = N_A$ particles. His rocket releases the reactants into the reaction chamber at 10 moles per second and 20 moles per second respectively. Given chamber pressure P_c , temperature T_a , no heat loss, and specific heat capacities c_{p1} and c_{p2} for H_2O and CO_2 respectively, solve the following problems.

- Find the initial velocity v_0 of the products within the reaction chamber of area A_1 .
- Bob measures the temperature of the exhaust exiting the nozzle as T_e . Find the final exhaust velocity v_e of the products as they exit the right end of the nozzle of area A_3 after passing through the throat with area A_2 where they reached the speed of sound.
- Bob happens to be in a vacuum with absolutely no gravity and decides to let the rocket burn all of its fuel starting at rest. Calculate the total efficiency of this rocket's engine over the course of its entire burn (the portion of the fuel's total initial energy transferred to the rocket's kinetic energy).

- D) If Bob starts the rocket at a greater initial speed, it will gain more kinetic energy. He can keep increasing this initial speed until rocket's gain in kinetic energy is greater than the fuel's chemical potential energy. Explain how conservation of energy is not violated. Include a verbal explanation followed by a separate mathematical explanation.

10. Protein Shake

Jayden Daniels² was making a protein shake, which included protein powder molecules of radius r and mass m . Interestingly, he noticed when the protein powder was soggy, it stuck to the side of a mixer with radius $R \gg r$. Consider the mixer to contain a fluid with coefficient of viscosity η rotating everywhere at a constant angular velocity ω . Assume there are $N \gg 1$ protein powder molecules stuck to the side of the mixer and the fluid has a surface tension γ and is filled to a height h . Estimate (excluding dimensionless coefficients) the minimum coefficient of friction μ necessary for this phenomena to occur.

11. Push Harder

A long cart moves with a speed $v \sim c$. Sand is dropped in cart at a rate $\frac{dm}{dt} = \sigma$ in the ground frame, and the sand is always dropped at the same place over the ground. Alice is stationary behind the cart, and pushes the back of the cart³ with a force F to keep it moving at a constant speed v .

- A) Compute F in the ground frame.
 B) By the longitudinal force transformation of special relativity, F must be the same for an observer in the cart's frame. Verify this explicitly by computing F in the cart's frame.

You may assume the cart is made of a maximally rigid material.

12. Windy Windmills

A wind turbine can only extract a certain fraction of the kinetic energy of the air - if it were to extract all, then the air would stop and no wind would go through the turbine - causing it to extract no energy. This fraction has a theoretical maximum - evaluate it (and show work). Assume air is incompressible.

13. Spaceships Galore

Two spaceships start to accelerate with a constant proper acceleration $a_1 = 15 \frac{m}{s^2}$ and $a_2 = 30 \frac{m}{s^2}$ respectively. Both spaceships motions are in a straight line with respect to their starting planet. However, in the first planet's frame, the second spaceship's direction is 10° from the first spaceships. In the first spaceship's frame, after a year, how far is the second spaceship?

²Greatest Quarterback in NFL history by the way

³You can assume the mass of Alice's arms are negligible